Elementary Real Analysis

Labix

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Contents

1 Properties of the Real Numbers

1.1 The Completeness Property

Recall that we constructed the real numbers $\mathbb{R}$ as Dedekind cuts of $\mathbb{Q}$. The real numbers are also equipped with addition and multiplication, and is totally ordered.

Definition 1.1.1 (Intervals)
Let $I \subseteq \mathbb{R}$ be a subset. We say that I is an interval if for all $a, b \in I$, we have $x \in I$ for all $a < x < b$.

We use the following convention:

- The open interval $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$
- The closed interval $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$
- $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$
- $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$

Definition 1.1.2 (Upper and Lower Bounds)
Let S be a non empty subset of $\mathbb{R}$
- S is said to be bounded above if there exists $u \in \mathbb{R}$, called an upper bound of S such that $x \leq u$ for all $x \in S$
- S is said to be bounded below if there exists $l \in \mathbb{R}$, called a lower bound of S such that $l \leq x$ for all $x \in S$
- S is said to be bounded if there exists a $M \in \mathbb{R}$ such that $|x| \leq M$ for all $x \in S$

Lemma 1.1.3
Let $S \subseteq \mathbb{R}$ be a subset. Then S is bounded if and only if it is bounded above and bounded below.

Proof
If S is bounded, then $-M \leq x \leq M$ thus it is bounded above and below. If S is bounded above by M and bounded below by N, then $|x| \leq \max(|M|, |N|)$. ■

Definition 1.1.4 (Supremum)
Let S be a subset of $\mathbb{R}$. We say that $U \in \mathbb{R}$ is the supremum of S when
- $x \leq U$ for all $x \in S$
- If u is an upper bound of S then $U \leq u$

We denote the supremum of S as $\sup(S) = U$

Definition 1.1.5 (Infinum)
Let S be a non-empty subset of $\mathbb{R}$. We say that $L \in \mathbb{R}$ is the infinum of S when
- $L \leq x$ for all $x \in S$
- If l is an upper bound of S then $l \leq L$

We denote the infinum of S as $\inf(S) = L$

Proposition 1.1.6
Let S be a non-empty subset of $\mathbb{R}$. Then

$$-\sup(S) = \inf(-S)$$

where $-S = \{-x | x \in S\}$.

Proof
Since for all $x \in S$ $x \leq \sup(S)$. But this implies that $-\sup(S) \leq -x$ for all $x \in S$ thus for all $-x \in -S$. Then by definition $-\sup(S) = \inf(-S)$. ■

Proposition 1.1.7 (Approximation Property)
Let S be a non-empty subset of $\mathbb{R}$.
- If $\sup(S)$ exists then for all $\varepsilon > 0$ there exists $x \in S$ such that $\sup(S) - \varepsilon < x \leq \sup(S)$.
- If $\inf(S)$ exists then for all $\varepsilon > 0$ there exists $x \in S$ such that $\inf(S) \leq x < \inf(S) + \varepsilon$.

Proof
Notice that since $\sup(S)$ is the supremum of S, we have that $\sup(S) - \varepsilon$ is not the supremum of S Thus there exists some element $x \in S$ such that $\sup(S) - \varepsilon < x$. Mirror the proof for the infinum version. ■

Theorem 1.1.8 (The Completeness Property of Real Numbers)
Let $S \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ that is bounded above. Then S has a supremum.

1.2 Density of the Real Numbers

Theorem 1.2.1 (Archimedean Property)
For any real numbers a and b with $a > 0$ there exists $n \in \mathbb{N}$ such that $na > b$.

Proof
Suppose that $na \leq b$ for all n. Then the set of all na, denoted by S has an uppper bound b, thus also has a least upeer bound, say u. We have that there exists some $na \in S$ such that $u - a < na \leq u$. But this implies that $u < (n+1)a \in S$, a contradiction. ■

Proposition 1.2.2 (Floor Function)
For each $x \in \mathbb{R}$ there exists a unique integer $\lfloor x \rfloor$ such that

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$$

Proof
Let $A = \{n \in \mathbb{Z} : n \leq x\}$. By the archimedean property there exists $k > -x$ and thus $-k < x$ with $-k \in A$ hence A is non-empty. Let $a = sup(A)$. We have some $n \in A$ such that $a - 1 < n \leq a \leq x$ and thus $a + 1 < n + 1 \leq a + 1 \leq x + 1$. From this we have $n \leq x$ as well as $n + 1$ not in A thus $x < n + 1$.

Now suppose that $m \leq x < m + 1$ and $n \leq x < n + 1$. From these we derive that $m \leq x < n + 1$ and $n \leq x < m + 1$. Thus $m - n < 1$ and $m - n > -1$ and $m - n \in \mathbb{Z}$. and we have $m - n = 0$ ■

Definition 1.2.3 (Square Root of a Real Number)
Let $c \in \mathbb{R}^+$. Define the square root of c to be

$$\sqrt{c} = \sup\{x \in \mathbb{R} \mid x \geq 0 \text{ and } x^2 \leq c\}$$

The number $\sqrt{c}$ exists because of the completeness property.

Lemma 1.2.4
Let $c \in \mathbb{R}^+$. Then we have $(\sqrt{c})^2 = c$.

Example 1.2.5
The number $\sqrt{2}$ is irrational.

Proof
Suppose for a contradiction that $\sqrt{2}$ is rational. Then there exists $p, q \in \mathbb{N}^+$ and $\gcd(p, q) = 1$ such that $\sqrt{2} = \frac{p}{q}$. Then we have $p = \sqrt{2}q$ which implies that $p^2 = 2q^2$. But this implies that p^2 is even, and hence p is even. So there exists $k \in \mathbb{N}$ such that $p = 2k$. Substituting gives $4k^2 = 2q^2$ and $2k^2 = q^2$. But this also implies that q^2 is even and hence q is even. This is a contradiction since 2 is both a factor of p and q, but we assumed that $\gcd(p, q) = 1$. ■

Theorem 1.2.6 (Density of the Real Numbers)
Let $a, b \in \mathbb{R}$. Then the following are true.

- There exists $x \in \mathbb{Q}$ such that $a < x < b$.
- There exists $y \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < y < b$.

Proof
Suppose $a, b \in \mathbb{R}$ and $a < b$. By the archimedean property we have that $n(b-a) > 1$ for some $n \in \mathbb{N}$. Let $m = \lfloor na \rfloor + 1$. We have $na < m \leq na + 1 < nb$. Thus $a < \frac{m}{n} < b$.

Suppose $a, b \in \mathbb{R}$ and $a < b$. By the archimedean property we have that $n(b-a) > 1$ for some $n \in \mathbb{N}$. Let $m = \lfloor na \rfloor + 1$. We have $na < m \leq na + 1 < nb$. Thus $a < \frac{m}{n} < b$. ■

2 Sequences

2.1 Sequences and Convergences

We begin our discussion we the definition and the first properties of sequences, as well as convergence.

Definition 2.1.1 (Sequences)
A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is a function $f:\mathbb{N}\to\mathbb{R}$ that assigns to each natural number a real number $f(n)=a_n$.

Definition 2.1.2 (Limits and Convergence)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$. Let $a\in\mathbb{R}$. We say that the sequence converges to the limit a if for all $\varepsilon>0$, there exists $N\in\mathbb{N}$ such that

$$n>N \implies |a_n-a|<\varepsilon$$

In this case we write $(a_n)_{n\in\mathbb{N}}\to a$.

Example 2.1.3
The sequence $(1/n)_{n\in\mathbb{N}}$ converges to 0.

Proof
Let $\varepsilon>0$. Choose $N>\frac{1}{\varepsilon}$. This is possible by the Archimedean property. For $n>N$, we have

$$\frac{1}{n}<\frac{1}{N}<\varepsilon$$

■

Proposition 2.1.4 (Uniqueness of Limits)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$. Let $a,b\in\mathbb{R}$. Suppose that $(a_n)_{n\in\mathbb{N}}$ converges to a and b. Then $a=b$.

Proof
Choose ε to be $\frac{|a-b|}{10}$. Then there exists N_1 and N_2 such that $|a_n-a|\varepsilon$ for all $n>N_1$ and $|a_n-b|<\varepsilon$ for all $n>N_2$ respectively. When $n>\max(N_1,N_2)$ we have both inequalies hold together. Then

$$\begin{aligned}|a-b|&\leq|a_n-a|+|a_n-b|\\&\leq\frac{|a-b|}{10}+\frac{|a-b|}{10}\\&=\frac{|a-b|}{5}\end{aligned}$$

Which is a contradiction. ■

Proposition 2.1.5
A convergent sequence in $\mathbb{R}$ is bounded.

Proof
Suppose that $a_n\to a$. Fixing ε to be any number larger than 0, we have that $|a_n-a|<\varepsilon \implies a-\varepsilon<a_n<a+\varepsilon$ for all n larger than some number N. At the same time, all the terms less than N are bounded by $M=\max(|a_1|,|a_2|,\dots,|a_N|)$. Thus if we take

$$C=\max(M,|a+\varepsilon|,|a-\varepsilon|)$$

C would bound all the terms of a_n. ■

Proposition 2.1.6
Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be sequences with $a_n \to a$ and $b_n \to b$. If $\exists N \in \mathbb{N}$ such that $a_n \leq b_n$ for all $n > N$, then $a \leq b$.

Proof
Suppose that $b < a$. Then we have $b < \frac{a+b}{2} < a$. Choosing $\varepsilon = \frac{a-b}{2}$, then whenever $n > N_1$, we have

$$b_n - b < \frac{a-b}{2}$$

Similarly, choose $\varepsilon = \frac{b-a}{2}$, then whenever $n > N_2$, we have

$$a_n - a < \frac{b-a}{2}$$

Then we have

$$b_n < \frac{a+b}{2} < a_n$$

which is a contradiction. ■

Proposition 2.1.7 (Sandwich Theorem)
Let $(a_n)_{n\in\mathbb{N}}$, $(b_n)_{n\in\mathbb{N}}$, $(c_n)_{n\in\mathbb{N}}$ be sequences in $\mathbb{R}$ such that

$$a_n \leq b_n \leq c_n$$

for all n larger than some $N \in \mathbb{N}$. If $a_n \to L$ and $c_n \to L$ then $b_n \to L$.

Proof
Suppose $a_n \to L$ and $c_n \to L$. Then choose an ε that is greater than 0. Then there exists N_1 and N_2 such that $|a_n - L| < \varepsilon$ whenever $n > N_1$, and $|c_n - L| < \varepsilon$ whenever $n > N_2$. Then whenever $N = \max(N_1, N-2)$, we have $-\varepsilon < a_n - L < b_n - L < c_n - L < \varepsilon$, and thus we have $|b_n - L| < \varepsilon$ whenever $n > N$. ■

Definition 2.1.8 (Divergence to Infinity)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence.

- We say that the sequence diverges to ∞ if for all $C \in \mathbb{R}^+$, there exists $N \in \mathbb{N}$ such that $a_n > C$ for all $n > N$.
- We say that the sequence diverges to $-\infty$ if for all $C \in \mathbb{R}^-$, there exists $N \in \mathbb{N}$ such that $a_n < C$ for all $n > N$.
- We say that the sequence diverges if it does not converge and does not diverge to $\pm\infty$.

Proposition 2.1.9
Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be sequences in $\mathbb{R}$. If $b_n \geq a_n$ for all n larger than some $N \in \mathbb{N}$ and $a_n \to \infty$ then $b_n \to \infty$.

Proof
Suppose $a_n \to \infty$. Then we have for every $C > 0$ there exists some $N \in \mathbb{N}$ such that $a_n > C$ whenever $n > N$. Then we have $b_n > a_n > C$ for all $n > N$ thus $b_n \to \infty$. ■

Proposition 2.1.10 (Algebra of Sequences)
Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be sequences in $\mathbb{R}$. Let $a_n \to a$ and $b_n \to b$. Then the following are true.

- Sum Rule: $sa_n + tb_n \to sa + tb$ for any $s, t \in \mathbb{R}$
- Product Rule: $a_n b_n \to ab$
- Quotient Rule: $\frac{a_n}{b_n} \to \frac{a}{b}$ if $b \neq 0$
- $|a_n| \to |a|$

Proof
Suppose that $(a_n) \to a$ and $(b_n) \to b$

- Choosing any $\frac{\varepsilon}{|s|} > 0$, we have that whenever $n > N_1$ for some N_1, $|a_n - a| < \frac{\varepsilon}{|s|}$, thus $|sa_n - sa| < \varepsilon$, thus $(sa_n) \to sa$. Similarly, $(tb_n) \to tb$. Now for every $\frac{\varepsilon}{2} > 0$, we have that whenever $n > \max(N_1, N_2)$, $|sa_n - sa|$ and $|tb_n - tb|$ whenever $n > \max(N_1, N_2)$. Then

$$\begin{aligned}|(sa_n + tb_n) - (sa + tb)| &\leq |sa_n - sa| + |tb_n - tb| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon\end{aligned}$$

  Thus we have the desired result.

- Choose $M \geq |a|$ such that $|b_n| \leq M$ for all n. This is possible since both sequecnes are bounded. Then there exists $N \in \mathbb{N}$ we have $|a_n - a| < \frac{\varepsilon}{2M}$ and $|b_n - b| < \frac{\varepsilon}{2M}$ for all $n > N$. We then have

$$\begin{aligned}|a_nb_n - ab| &= |(a_n - a)b_n + a(b_n - b)| \\ &\leq |a_n - a||b_n| + |a||b_n - b| \\ &\leq M|a_n - a| + M|b_n - b| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon\end{aligned}$$

  Thus we have the desired result.

- We want to show that $\frac{1}{b_n} \to \frac{1}{b}$ since we can apply the product rule to produce the quotient rule. From the product rule we have that $bb_n \to b^2$. Choosing $\varepsilon = \frac{b^2}{2} > 0$, we have a fixed N such that $\left|bb_n - b^2\right| < \frac{b^2}{2}$ whenever $n > N$. Thus we have $\frac{b^2}{2} < bb_n$ whenever $n > N$. Then we have

$$\begin{aligned}\frac{b^2}{2} < |bb_n| &\implies \frac{1}{|bb_n|} < \frac{2}{b^2} \\ &\implies \left|\frac{b_n - b}{bb_n}\right| < \frac{2|b_n - b|}{b^2} \\ &\implies \left|\frac{1}{b_n} - \frac{1}{b}\right| < \frac{2|b_n - b|}{b^2}\end{aligned}$$

  Since we have $0 < \left|\frac{1}{b_n} - \frac{1}{b}\right| < \frac{2|b_n-b|}{b^2}$ then by sandwich theorem, we have that $\frac{1}{b_n} \to \frac{1}{b}$. Then by product rule, we have the desired result.

- For every $\varepsilon > 0$, there exists some N such that $||a_n| - |a|| \leq |a_n - a| < \varepsilon$ whenever $n > N$. Thus trivially $|a_n| \to a$.

■

Example 2.1.11
We have

$$\left(\frac{\sqrt{9n^6 + 4n^2 + 7}}{4n^3 + 2n + 5}\right)_{n\in\mathbb{N}} \to \frac{3}{4}$$

Proposition 2.1.12
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Then the following are true.

- If $a_n \to \infty$ then $\frac{1}{a_n} \to 0$
- If $a_n \to 0$ and there exists $N \in \mathbb{N}$ such that $a_n > 0$ for all $n > N$, then $\frac{1}{a_n} \to \infty$

Proof
We prove the following with care.

- Since $a_n \to \infty$, we have for every $C > 0$ there exists some N such that $a_n > C$ whenever $n > N$. Then we have for every $\varepsilon = \frac{1}{C} > 0$ there exists N such that $\left|\frac{1}{a_n}\right| < \frac{1}{C} = \varepsilon$. Thus we have the desired result.
- We have that for every $\varepsilon > 0$, there exists some N_1 such that $|a_n| < \varepsilon$ whenever $n > N_1$, also we have $a_n > 0$ when $n > N_2$. Choosing $C = \frac{1}{\varepsilon}$, then for all $n > \max(N_1, N_2)$, we have $|a_n| < \varepsilon \implies \left|\frac{1}{\varepsilon}\right| > \frac{1}{\varepsilon} = C$ Thus we have the desired result.

■

Definition 2.1.13 (Monotone Sequences)
A sequence (a_n) in $\mathbb{R}$ may have different properties as below.
- $(a_n)_{n\in\mathbb{N}}$ is strictly increasing if $a_n < a_{n+1}$ for all $n \in \mathbb{N}$
- $(a_n)_{n\in\mathbb{N}}$ is increasing if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$
- $(a_n)_{n\in\mathbb{N}}$ is strictly decreasing if $a_n > a_{n+1}$ for all $n \in \mathbb{N}$
- $(a_n)_{n\in\mathbb{N}}$ is decreasing if $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$
- $(a_n)_{n\in\mathbb{N}}$ is monotone if it is either increasing or decreasing

Theorem 2.1.14 (Monotone Sequence Theorem)
Let $(a_n)_{n\in\mathbb{N}}$ be a strictly increasing sequence and bounded above. Then $(a_n)_{n\in\mathbb{N}}$ converges.

Proof
Suppose $(a_n)_{n\in\mathbb{N}}$ is increasing and bounded. Then $S = \{a_n | n \in \mathbb{N}\}$ is bounded thus has a supremum. We have that

$$\sup(S) - \varepsilon < a_N \leq a_{N+1} \leq \cdots \leq \sup(S) < \sup(S) + \varepsilon$$

for all n larger than some $N \in \mathbb{N}$. Thus we have the for all $n > N$,

$$\sup(S) - \varepsilon < a_n < \sup(S) + \varepsilon \implies |a_n - \sup(S)| < \varepsilon$$

The proof is similar for the decreasing version. ■

Example 2.1.15
Let $r > 0$. Let $(a_n)_{n\in\mathbb{N}}$ be the sequence defined by $a_0 = \sqrt{r}$ and $a_n = \sqrt{r + a_{n-1}}$. Then we have

$$(a_n)_{n\in\mathbb{N}} \to \frac{1 + \sqrt{1 + 4r}}{2}$$

Proof
Let $L = \frac{1+\sqrt{1+4r}}{2}$. Notice that $L^2 - L - r = 0$. The sequence is strictly increasing since $\frac{a_n}{a_{n-1}} = \sqrt{\frac{r}{a_{n-1}} + 1} > 1$ and $a_n > 0$ by induction. We prove that the sequence is bounded above by L. When $n = 0$, we have $a_0 = \sqrt{r} = \sqrt{L^2 - L} \leq \sqrt{L^2} = L$. Suppose that $a_{n-1} \leq L$. Then we have

$$a_n = \sqrt{r + a_{n-1}} \leq \sqrt{r + L} = \sqrt{L^2} = L$$

Hence $(a_n)_{n\in\mathbb{N}}$ is strictly increasing and bounded above. By the monotone sequence theorem, we conclude that $(a_n)_{n\in\mathbb{N}}$ is convergent. Suppose that a is the limit. Then since $(a_{n+1})_{n\in\mathbb{N}}$ and $(a_n)_{n\in\mathbb{N}}$ converges to the same limit, taking limit on the equation $a_{n+1}^2 = r + a_n$, we get

$$a^2 = r + a$$

by the algebra of sequences. Then we have $a^2 - a - r = 0$. Solving gives $a = L$ since the other root is negative, but $(a_n)_{n\in\mathbb{N}}$ is strictly increasing and positive. ■

2.2 Subsequences

Definition 2.2.1 (Subsequences)
A subsequence of a sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is a sequence $(a_{n_k})_{k\in\mathbb{N}}$, where

$$0 \leq n_1 < n_2 < \dots$$

Proposition 2.2.2
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$ and let $a_n \to a$. Then any subsequence of $(a_{n_k})_{k\in\mathbb{N}}$ converges to a.

Proof
Given $\varepsilon > 0$ there exists some N such that $|a_n - a| < \varepsilon$ whenever $n > N$. Since $n_k > k$, we have that $|a_{n_k} - a| < \varepsilon$ for all $k > N$, thus we have $a_{n_k} \to a$ ■

Theorem 2.2.3 (Bolzano-Weierstrass Theorem)
Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Proof
Let $(a_n)_{n\in\mathbb{N}}$ be a bounded sequence, say $c_0 \leq a_n \leq d_0$ for all n. Bisect the interval $I_0 = [c_0, d_0]$. Observe that if the union of two sets contains infinitely many terms, than at least one of the sets must contain infinitely many terms of the sequence.

Suppose I_1 is the said set. From I_1, choose one such term, say a_{n_1}. Bisect I_1 again and call I_2 the interval with infinitely many terms of the sequence. Choose one such term say a_{n_2}, with $n_2 > n_1$, by repeating this procedure, we obtain a subsequence (a_{n_k}) and a sequence of intervals $I_k = [c_k, d_k]$ such that

$$c_0 \leq c_{k-1} \leq c_k \leq a_{n_k} \leq d_k \leq d_{k-1} \leq d_0$$

and $d_{k+1} - c_{k+1} = \frac{1}{2}(d_k - c_k)$.

Observe that (c_k) and (d_k) are monotonic and bounded, thus converges to c and d respectively. Since

$$d_k - c_k = 2^{-k}(d_0 - c_0) \to 0$$

we have that $c = d$. By the sandwich theorem, we have $a_{n_k} \to c$ ■

Definition 2.2.4 (Limit Superior and Limit Inferior)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Define the limit superior of the sequence to be the the limit of

$$s_n = \sup_{m\geq n} a_m$$

if it exists.

Define the limit inferior of the sequence to be the limit of

$$t_n = \inf_{m\geq n} a_m$$

if it exists.

Proposition 2.2.5
Let $(a_n)_{n\in\mathbb{N}}$ be a bounded sequence. Then s_n and t_n converges.

Proof
Trivially we must have s_n a decreasing sequence and t_n an increasing sequence since $\{a_m | m \geq n_2\} \subset \{a_m | m \geq n_1\}$ as long as $n_1 < n_2$. Since $(a_n)_{n\in\mathbb{N}}$ is bounded, so is the subsequences s_n and t_n. Thus s_n and t_n is convergent by the monotone convergent theorem. ■

Proposition 2.2.6
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Then $a_n \to a$ if and only if

$$\lim_{n\to\infty} \sup_{m\geq n} a_m = \lim_{n\to\infty} \inf_{m\geq n} a_m = a$$

Proof
Firstly suppose that $a_n \to a$. Then since $(a_n)_{n\in\mathbb{N}}$ is convergent it is bounded. Then by the above theorem the subsequences converges. By theorem 2.2.2 we must have the limit suprerior and limit inferior converge to a.

Now suppose that the limit superior and limit inferior both converges to $a \in \mathbb{R}$. Then this means that fixing $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that $a - \varepsilon < \sup_{m\geq n} a_m < a + \varepsilon$ for all $n > N_1$. Similarly there exists N_2 such that $a - \varepsilon < \inf_{m\geq n} a_m < a + \varepsilon$ for all $n > N_2$. This means that

$$a - \varepsilon < \inf_{m\geq n} a_m < a_m < \sup_{m\geq n} a_m < a + \varepsilon$$

for all $m \geq n > \max N_1, N_2$. Thus $a_n \to a$ and we are done. ■

2.3 Cauchy Sequences

Definition 2.3.1 (Cauchy Sequence)
A sequence $(a_n)_{n\in\mathbb{N}}$ is said to be Cauchy if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n, m > N \implies |a_n - a_m| < \varepsilon$$

Proposition 2.3.2
Every convergent sequence is Cauchy.

Proof
Suppose that $(a_n)_{n\in\mathbb{N}}$ is convergent. Choose $\frac{\varepsilon}{2} > 0$, then there exists N such that $|a_n - a| < \frac{\varepsilon}{2}$ and $|a_m - a| < \frac{\varepsilon}{2}$ whenever $n, m > N$. Then

$$\begin{aligned} |a_n - a_m| &\leq |a_n - a| + |a_m - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

Thus we have the desired result. ■

Proposition 2.3.3
Every Cauchy Sequence is bounded.

Proof
Suppose that $(a_n)_{n\in\mathbb{N}}$ is cauchy. Then fixing $\varepsilon > 0$ there exists some N such that $|a_n - a_m| < \varepsilon$ whenever $n, m > N$. Then we have

$$\begin{aligned} |a_n| &\leq |a_n - a_N| + |a_N| \\ &< |a_N| + \varepsilon \end{aligned}$$

whenever $n > N$. Thus we have that a_n is bounded whenever $n > N$. For $n \leq N$, it is bounded by $M = \max(a_1, a_2, \dots, a_N)$. So by taking the max of M and $a_N + \varepsilon$, we have that all terms in the sequence are bounded. ■

Proposition 2.3.4
Every Cauchy sequence is convergent.

Proof
Since a cauchy sequence is bounded, we have from the Bolzano Weierstrass theorem that there is subsequence, say $(a_{n_k})_{k\in\mathbb{N}}$ that converges to say, a. Then fixing $\frac{\varepsilon}{2} > 0$, there exists some $N_1 \in \mathbb{N}$ such that $|a_{n_k} - a| < \frac{\varepsilon}{2}$ whenever $k > N_1$. Also since it is cauchy, we have that when we have $\frac{\varepsilon}{2} > 0$ there exists some $N_2 \in \mathbb{N}$ such that $|a_n - a_{n_k}| < \frac{\varepsilon}{2}$ whenever $n, k > N_2$. Choosing $N = \max(N_1, N_2)$, then we have

$$\begin{aligned}|a_n - a| &\leq |a_n - a_{n_k}| + |a_{n_k} - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon\end{aligned}$$

Thus we have the desired result. ■

Example 2.3.5 (Geometric Sequences)
Let $x \in \mathbb{R}$ be a real number. Then the following is true.

$$(x^n)_{n\in\mathbb{N}} \to \begin{cases} \infty & \text{if } x > 1 \\ 1 & \text{if } x = 1 \\ 0 & \text{if } -1 < x < 1 \\ \text{diverges} & \text{if } x \leq -1 \end{cases}$$

Proof
We treat different cases separately.

- We have $x^n \geq (1 + n(x-1))$ by the Bernoulli's Inequality. Since $(1 + n(x-1))$ diverges to $+\infty$, we also have that (x^n) diverges to $+\infty$.
- Now when $x = 1$ we have the sequence $1, 1, 1, \dots$ thus it converges to 1.
- If $|x| < 1$, let $x = \frac{1}{1+t}$ with $t > 0$. Then by Bernoulli's Inequality, we have $(1+t)^n \geq 1 + nt > nt$, thus $\frac{1}{(1+t)^n} \leq \frac{1}{1+nt} < \frac{1}{nt}$. But since $nt \to \infty$, we have that $\frac{1}{nt} \to 0$. Then for every $\varepsilon > 0$ there exists N such that $\left|\frac{1}{nt}\right| < \varepsilon$ whenever $n > N$. Using this, we have $|x^n| = \left|\frac{1}{(1+t)^n}\right| < \varepsilon$. Thus we have the desired result.
- It is easy to show that it neither goes to $\pm\infty$ nor converges.

■

Definition 2.3.6 (Strictly Contracting)
A sequence $(a_n)_{n\in\mathbb{N}}$ is strictly contracting if $|a_{n+2} - a_{n+1}| \leq l|a_{n+1} - a_n|$ for all $n \in \mathbb{N}$ where $l \in (0, 1)$

Proposition 2.3.7
Every strictly contracting sequence is Cauchy.

Proof
Suppose that $(a_n)_{n\in\mathbb{N}}$ is a strictly contracting sequence. Without loss of generality suppose $n > m$.

$$\begin{aligned}|a_n - a_m| &\leq |a_n - a_{n-1}| + |a_{n_1} - a_{n-2}| + \cdots + |a_{m+2} - a_{m+1}| + |a_{m+1} - a_m| \\ &\leq l^{n-m}|a_{m+1} - am| + l^{n-m-1}|a_{m+1} - am| + \cdots + l|a_{m+1} - a_m| + |a_{m+1} - a_m| \\ &= (l^{n-m} + l^{n-m-1} + \cdots + l + 1)|a_{m+1} - a_m| \\ &= \frac{1 - l^{n-m+1}}{1 - l}|a_{m+1} - a_m| \\ &= \frac{1 - l^{n-m+1}}{1 - l}l^m|a_2 - a_1| \\ &\leq \frac{l^m}{1 - l}|a_2 - a_1|\end{aligned}$$

As $m \to \infty$ we have $\left(\frac{l^m}{1-l}\right) \to 0$ by the geometric sequence. Then for every $\frac{\varepsilon}{|a_2-a_1|}$, there exists N such that $\left|\frac{l^m}{1-l}\right| < \frac{\varepsilon}{|a_2-a_1|}$ whenever $n > N$. Using this, we have that $|a_n - a_m| \leq \frac{\varepsilon}{|a_2-a_1|} < \varepsilon$. Thus we have the desired result. ■

Proposition 2.3.8 (nth roots)
For every $x \in \mathbb{R}$ and $n \in \mathbb{N}$ there exists a unique nth root denoted by $x^{\frac{1}{n}}$.

Proof
Let $A = \{x > 0 : x^n > a\}$ We want to show that the infinum of this set is exactly the nth root of a. Note that $(1 + a)^n \geq 1 + na > a$ thus A is non-empty. Also since $x > 0$ we have that 0 is a lower bound and thus its infinum exists by the completeness axiom. We let $b = \inf(A)$. Since b is the infinum, we have some $a_k \in A$ such that $b \leq a_k < b + \frac{1}{k}$. By the sandwich theorem we have $a_k \to b$. By the product rule of series, we have $a_k^n \to b^n$. Recall that $a_k \in A$, then we have $a_k^n > a$, and $\lim_{k\to\infty} a_k^n \geq a$, thus $b^n \geq a$.

Assume now that $b^n > a$, we have $0 < \frac{a}{b^n} < 1$ so we may choose $\delta > 0$ so that $\delta < \frac{b}{n}(1 - \frac{a}{b^n})$. We now show that $b - \delta \in A$. Not that from our definition of δ, we have $\delta < \frac{b}{n}(1 - \frac{a}{b^n})$ and since $n > 1$ and $0 < 1 - \frac{a}{b^n} < 1$, we have $\delta < \frac{b}{n}\left(1 - \frac{a}{b^n}\right) < b$. Thus we have $\frac{\delta}{b} < 1$, and $-\frac{\delta}{b} > -1$. Thus we can apply Bernoulli's Inequality.

$$\begin{aligned}(b - \delta)^n &= b^n\left(1 - \frac{\delta}{b}\right)^n \\ &\geq b^n\left(1 - n\frac{\delta}{b}\right) && \text{(By Bernoulli's Inequality)} \\ &> a\end{aligned}$$

since $\delta < \frac{b}{n}\left(1 - \frac{a}{b^n}\right) \implies b^n\left(1 - n\frac{\delta}{b}\right) > a$. This proves that $b - \delta \in A$, contradicting the fact that b is the infinum. Hence we can only have $b^n = a$ This completes the proof of existence of the nth root.

Now we prove the uniqueness of the nth root. Suppose that $b^n = c^n = a$. Without loss of generality assume $b \leq c$We have

$$\begin{aligned}0 &= c^n - b^n \\ 0 &= (c - b)(c^{n-1} + bc^{n-2} + \cdots + b^{n-2}c + b^{n-1})\end{aligned}$$

But $c^{n-1} + bc^{n-2} + \cdots + b^{n-2}c + b^{n-1} > nb^{n-1}$. Thus we can only have $b = c$. ■

Example 2.3.9
Let $x > 0$ be a real number. Then we have

$$\left(x^{\frac{1}{n}}\right)_{n\in\mathbb{N}} \to 1$$

Proof
Consider the case with $x \geq 1$. We have $x^{\frac{1}{n}} \geq 1$. From the Bernoulli's Inequality, we have $\left(1 + x^{\frac{1}{n}} - 1\right) \geq 1 + n\left(x^{\frac{1}{n}} - 1\right) \implies x \geq 1 + n\left(x^{\frac{1}{n}} - 1\right)$. Thus we have $0 < x^{\frac{1}{n}} - 1 \leq \frac{x-1}{n}$. Since $\left(\frac{x-1}{n}\right) \to 0$, we have that $\left(x^{\frac{1}{n}} - 1\right) \to 0$ from the sandwich theorem and thus $\left(x^{\frac{1}{n}}\right) \to 1$. For the case $0 < x < 1$, note that $\left(\frac{1}{x^{\frac{1}{n}}}\right) \to 1$ since $\frac{1}{x} > 1$. Using the algebra for sequences, we have $\left(\frac{1}{\frac{1}{x}^{\frac{1}{n}}}\right) = \left(x^{\frac{1}{n}}\right) \to 1$. ■

Example 2.3.10
The sequence $n^{\frac{1}{n}}$ converges to 1.

Proof
Note that $n \geq 1$ and $n^{\frac{1}{2n}} > 0$, we have

$$\begin{aligned}\sqrt{n} &= (1 + (n^{\frac{1}{2n}} - 1))^n \\ &\geq 1 + n(n^{\frac{1}{2n}} - 1) \\ &> n(n^{\frac{1}{2n}} - 1)\end{aligned}$$

from Bernoulli's Inquality. Thus we have that $1 \leq n^{\frac{1}{2n}} < \frac{1}{\sqrt{n}} + 1$. So by the sand wich theorem we have $(n^{\frac{1}{2n}}) \to 1$ and $(n^{\frac{1}{n}}) = (n^{\frac{1}{2n}})^2 \to 1$ by the arithmetic of sequences. ■

2.4 Tests for Convergence

Theorem 2.4.1 (Ratio Lemma)
Suppose $0 \leq l < 1$. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence.
- If there exists some $N \in \mathbb{N}$ $\frac{a_{n+1}}{a_n} \leq l$ for all $n > N$, then $a_n \to 0$
- If $\frac{a_{n+1}}{a_n} \to l$ then $a_n \to 0$

Proof
Suppose that $0 \leq l < 1$

- Note that since $a_n \leq la_{n-1}$, we have $a_n \leq l^n a_N$. Since $0 < a_n \leq l^n a_N$, and $(l^n) \to 0$ by geometric sequences, we have $a_n \to 0$.
- Since $\frac{a_{n+1}}{a_n} \to l$, we have that for every $\varepsilon > 0$, there exists N such that $\left|\frac{a_{n+1}}{a_n} - l\right| < \varepsilon$ whenever $n > N$. Then choosing ε such that $l + \varepsilon < 1$, we have that all the terms after a_N being less than $l + \varepsilon < 1$, thus apply the ratio lemma to have $a_n \to 0$.

■

Example 2.4.2
Let $x \in \mathbb{R}$ be a real number. Let $k \in \mathbb{N}$ be fixed. Then we have

$$\left(\frac{x^n}{n^k}\right)_{n\in\mathbb{N}} \to \begin{cases} \infty & \text{if } x > 1 \\ 0 & \text{if } 0 < x \leq 1 \end{cases}$$

Proof
Consider $\frac{a_{n+1}}{a_n}$ with $a_n = \left(\frac{x^n}{n^k}\right)$, We have that

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{x^{n+1}}{(n+1)^k}\frac{n^k}{x^n} \\ &= \left(1 - \frac{1}{n+1}\right)^k x\end{aligned}$$

We have that $\left(1 - \frac{1}{n+1}^k\right) \to 0$ so by the ratio lemma, it converges to 0 whenever $0 < x \leq 1$. Then suppose that $b_n = \frac{1}{a_n}$. Then $\frac{b_{n+1}}{b_n} = \frac{1}{(1-\frac{1}{n+1})^k}\frac{1}{x}$ thus b_n tends to 0 whenever $x > 1$. Then by taking the reciprocal we have $a_n \to \infty$ when $x > 1$. ■

2.5 Decimal Representations of Real Numbers

Definition 2.5.1 (Decimal Representation)
Let $x \in \mathbb{R}$ be a real number. A decimal representation of x is a sequence $(d_n)_{n\in\mathbb{N}}$ such that

$$\left(\sum_{k=0}^{n}\frac{d_k}{10^k}\right)_{n\in\mathbb{N}} \to x$$

where $d_0 \in \mathbb{Z}$ and $d_n \in \{0, \dots, 9\}$ for all $n \geq 1$. We write $x = d_0, d_1 d_2 \dots$ to represent x in this decimal representation.

Lemma 2.5.2
Let $x \in \mathbb{R}$ be a real number. Then a decimal representation of x exists.

Proof
Let $d_0 = \lfloor x \rfloor$. For each $n \in \mathbb{N} \setminus \{0\}$, define $d_n \in \mathbb{N}$ to be the largest number in $\{0, \dots, 9\}$ such that $\sum_{k=0}^{n}\frac{d_k}{10^k} \leq x$. Since $d_0 \leq \lfloor x \rfloor$, this is always possible. Define $S_n = \sum_{k=0}^{n}\frac{d_k}{10^k}$. By construction, $(S_n)_{n\in\mathbb{N}}$ is an increasing sequence and bounded above by x.

Let $M < x$. Suppose for a contradiction that $S_n < M$ for all $n \in \mathbb{N}$. ■

Decimal representations are far from unique.

Example 2.5.3
We have

$$1 = 0.\overline{9}$$

Definition 2.5.4 (Categorization of Decimals)
Let $x \in \mathbb{R}$ be a real number. Let $x = d_0.d_1 d_2 \dots$ be a decimal representation of x. We say that the decimal is

- Terminating if there exists $N \in \mathbb{N}$ such that $d_n = 0$ for all $n > N$.
- Recurring if there exists $N, r \in \mathbb{N}$ such that $d_n = d_{n+r}$ for all $n > N$.
- Non-recurring if it is neither terminating nor recurring.

Proposition 2.5.5
Let $x \in \mathbb{R}$. Then x is irrational if and only if x is non-recurring.

Proof
Suppose that $x = \frac{p}{q}$ with $p, q \in \mathbb{N}$. To represent a negative real number simply append the minus operation in front of $\frac{p}{q}$. Now, remove the integer part of x so that the new number, y, is between

0 and 1. Then y is also a rational number since alegra under rationals are closed. Let $y = \frac{m}{n}$. Let $m = d_0 n + r_1$. Then let $10r_1 = d_1 n + r_2$. Recursively define $10r_{k-1} = d_{k-1}n + r_k$. Observe that $r_k \in \{o, \dots, n-1\}$. Thus the operation eventually repeats, and we obtain a recurring decimal. If the recurring block of the decimal is only 0, then it is in fact a terminating decimal, a subset of recurring decimals. Note that the proof here required knowledge on basic number theory.

Now suppose that we have a recursive decimal x with $0 < x < 1$, a non-recurring block of length p and recurring block of length q. Then we have $(10^{p+q} - 1)x = n \in \mathbb{N}$. Then we have $x = \frac{n}{10^{p+q}-1}$. Append any integer to the front of the decimal then any recurring decimal can be represented by a rational number. ■

3 Limits and Continuity

3.1 Limits at a Point

Definition 3.1.1 (Limits at a Point)
Let $f:(a,c)\cup(c,b)\to\mathbb{R}$ be a function. We say that f converges to $L\in\mathbb{R}$ at c if for every $\varepsilon>0$, there exists $\delta>0$ such that for all x for which $0<|x-c|<\delta$, we have $|f(x)-L|<\varepsilon$. In this case, we write

$$\lim_{x\to c}f(x)=L$$

Definition 3.1.2 (One Sided Limits)
Let $f:(a,c)\cup(c,b)\to\mathbb{R}$ be a function.

- We say that
$$\lim_{x\to c^+}f(x)=L\in\mathbb{R}$$
if for every $\varepsilon>0$ there exists $\delta>0$ such that for all $x\in(c,c+\delta)$ we have $|f(x)-L|<\varepsilon$
- We say that
$$\lim_{x\to c^-}f(x)=L\in\mathbb{R}$$
if for every $\varepsilon>0$ there exists $\delta>0$ such that for all $x\in(c-\delta,c)$ we have $|f(x)-L|<\varepsilon$

Lemma 3.1.3
Let $f:(a,c)\cup(c,b)\to\mathbb{R}$ be a function. Let $L\in\mathbb{R}$. Then $\lim_{x\to c}f(x)=L$ if and only if

$$\lim_{x\to c^+}f(x)=\lim_{x\to c^-}f(x)=L$$

Proof
An easy manipulation of the definition of limits. ■

Proposition 3.1.4
Let $f:(a,c)\cup(c,b)\to\mathbb{R}$ be a function. Then

$$\lim_{x\to c}f(x)=L$$

if and only if for every sequence (x_n) in $(a,b)\setminus\{c\}$ with $(x_n)\to c$ we have $(f(x_n))\to L$.

Proof
Suppose that $\lim_{x\to c}f(x)=L$. Then for every $\varepsilon>0$ there exists a number $\delta>0$ such that if $|x-c|<\delta$ then $|f(x)-L|<\varepsilon$. Consider a sequence with $(x_n)\to c$. Then for every $\delta>0$ we have that $|x_n-c|<\delta$ for all n larger than some N. However this implies that $|f(x_n)-L|<\varepsilon$ for all n larger than N. Thus we have $(f(x_n))\to L$.

Now suppose that $(x_n)\to c \implies (f(x))\to L$. This means that for all $\delta>0$, there exists N such that $|x_n-c|<\delta$ for all $n>N$. Consider any x in the interval I. Then there is a sequence (x_n) that contains x. That sequence has the properties that $|x_n-c|<\delta$. We thus have $|f(x)-L|<\varepsilon$ by assumption. ■

Proposition 3.1.5 (Algebra of Limits)
Let $f,g:(a,c)\cup(c,b)\to\mathbb{R}$ be functions. Let $\lim_{x\to c}f(x)=L$ and $\lim_{x\to c}g(x)=L$. Then the following are true.

- $\lim_{x\to c}(f(x)+g(x))=L+M$
- $\lim_{x\to c}(f(x)g(x))=LM$
- If $M\neq 0$, then $\lim_{x\to c}(f(x)/g(x))=L/M$

Proof
Convert them to sequential definition of limits and apply algebra of sequences. ■

Theorem 3.1.6 (Sandwich Theorem for Limits)
Let $f, g, h : (a, c) \cup (c, b) \to \mathbb{R}$ be functions. If $f(x) \leq h(x) \leq g(x)$ for all $x \in (a, c) \cup (c, b)$, and

$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = L$$

then we have

$$\lim_{x\to c} h(x) = L$$

Proof
Convert them to sequential definition and apply the sandwich theorem of sequences. ■

Definition 3.1.7 (Limits to Infinity)
Let $f : (a, c) \cup (c, b) \to \mathbb{R}$ be a function.

- We say that f converges to ∞ as $x \to c$ if for all $M > 0$ there exists $\delta > 0$ such that for all x for which $0 < |x - c| < \delta$, we have $f(x) > M$. In this case we write

$$\lim_{x\to c} f(x) = \infty$$

- We say that f converges to $-\infty$ as $x \to c$ if for all $M < 0$ there exists $\delta > 0$ such that for all x for which $0 < |x - c| < \delta$, we have $f(x) < M$. In this case we write

$$\lim_{x\to c} f(x) = -\infty$$

Definition 3.1.8 (Limits at Infinity)
Let $f : \mathbb{R} \to \mathbb{R}$ be a function.

- We say that f converges to $L \in \mathbb{R}$ at positive infinity if for every $\varepsilon > 0$ there exists a $M > 0$ such that $x > M$ implies $|f(x) - L| < \varepsilon$. In this case we write

$$\lim_{x\to\infty} f(x) = L$$

- We say that f converges to $L \in \mathbb{R}$ at negative infinity if for every $\varepsilon > 0$ there exists a $M < 0$ such that $x < M$ implies $|f(x) - L| < \varepsilon$ In this case we write

$$\lim_{x\to-\infty} f(x) = L$$

3.2 Continuity

Definition 3.2.1 (Continuity)
A function $f : I \subseteq \mathbb{R} \to \mathbb{R}$ is said to be continuous at $c \in I$ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $x \in I$,

$$|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon$$

Example 3.2.2
Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by $f(x) = x$. Then f is continuous at all $c \in \mathbb{R}$.

Proof
Let $\varepsilon > 0$. Choose $\delta = \varepsilon$. For any $x \in \mathbb{R}$ such that $0 < |x - c| < \delta$, we have

$$|f(x) - f(c)| = |x - c| < \delta = \varepsilon$$

■

Definition 3.2.3 (Sequential Continuity)
Let $f : (a, b) \to \mathbb{R}$ be a function. Let $c \in (a, b)$. We say that f is sequentially continuous at c if for

every sequence $(x_n)_{n\in\mathbb{N}} \subset (a,b)$ that converges to c, then

$$(f(x_n))_{n\in\mathbb{N}} \to f(c)$$

Proposition 3.2.4
Let $f : (a,b) \to \mathbb{R}$ be a function. Let $c \in (a,b)$. Then the following are equivalent.

- f is continuous at c.
- f is sequentially continuous at c.
- $\lim_{x\to c} f(x) = f(c)$.

Proof
We first prove the forward implication. Suppose that f is continuous at c and $(x_n) \to c$. Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in I$, $|x-c| < \delta \implies |f(x) - f(c)| < \varepsilon$. Now, choose N so that if $n > N$, then $|x_n - c| < \delta$. Then if $n > N$ we have $|f(x_n) - f(c)| < \varepsilon$. Thus $(f(x_n)) \to c$.

Now suppose f is not continuous at c. Then there are some $\varepsilon > 0$ such that there exists some x with $|x-c| < \delta \implies |f(x) - f(c)| \geq \varepsilon$. We now construct a sequence as follows. For each n, choose x_n such that $|x_n - c| < \delta = \frac{1}{n}$ but $|f(x_n) - f(c)| \geq \varepsilon$. Then the sequence constructed tends to c since $\left(\frac{1}{n}\right) \to 0$ but $f(x_n)$ does not converge to $f(c)$.

f is continuous if and only if for all $\epsilon > 0$, there exists $\delta > 0$ such that $x \in (c-\delta, c+\delta)$ implies $|f(x) - f(c)| < \epsilon$. But this is precisely the definition of the limit of f. ∎

Example 3.2.5
Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ x & \text{otherwise} \end{cases}$$

Then f is continuous at $c \in \mathbb{R}$ if and only if $c = 0$.

Proof
Suppose that $c = 0$. Let $\varepsilon > 0$. Let $\delta = \varepsilon$. Suppose that $x \in \mathbb{R}$ such that $0 < |x| < \delta$. Then $|f(x) - f(0)| = |f(x)|$. There are two cases. If $x \in \mathbb{Q}$, then $|f(x)| = 0$ so that $|f(x)| < \delta = \varepsilon$. If $x \in \mathbb{R} \setminus \mathbb{Q}$, then $|f(x)| = |x| < \delta = \varepsilon$. Hence f is continuous at 0.

Now suppose that $c \neq 0$. There are two cases. Suppose that $c \in \mathbb{Q}$. For each $n \in \mathbb{N}^+$, choose $a_n \in \mathbb{R} \setminus Q$ such that $c - \frac{1}{n} < a_n < c$. This is possible since irrational numbers are dense in $\mathbb{R}$. Then by sandwich theorem, we conclude that $(a_n)_{n\in\mathbb{N}} \to c$. But $(f(a_n))_{n\in\mathbb{N}} = (a_n)_{n\in\mathbb{N}} \to c$ but $f(c) = 0$. Hence by the sequential continuity characterization, f is not continuous at c. Finally suppose that $c \in \mathbb{R} \setminus \mathbb{Q}$. For each $n \in \mathbb{N}^+$, choose $b_n \in \mathbb{Q}$ such that $c - \frac{1}{n} < b_n < c$. This is possible since rational numbers are dense in $\mathbb{R}$. Then by sandwich theorem, we conclude that $(b_n)_{n\in\mathbb{N}} \to c$. But $(f(b_n))_{n\in\mathbb{N}} = (0)_{n\in\mathbb{N}} \to 0$ but $f(c) = c$. Hence by the sequential continuity characterization, f is not continuous at c. ∎

Proposition 3.2.6
Let $f, g : I \subseteq \mathbb{R} \to \mathbb{R}$ and continuous at $c \in I$. Then

- $f + g$ is continuous at c
- $f \cdot g$ is continuous at c
- If $g(c) \neq 0$ then $\frac{f}{g}$ is continuous at c

Proof
The three proofs are simple. Using sequential continuity, since we have the algebra of sequences,

we also conclude the algebra of continuous functions. ■

Proposition 3.2.7
Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ and $g : J \subseteq \mathbb{R} \to I$. If f is continuous at $g(c)$ and g is continuous at c, then $f \circ g$ is continuous at c.

Proof
Let $x_n \to c$, then $g(x_n) \to g(c)$ thus $f(g(x_n)) \to f(g(c))$. ■

Theorem 3.2.8 (Intermediate Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous and suppose that $f(a) < u < f(b)$. Then

$$f(c) = u$$

for some $c \in (a, b)$.

Proof
Consider the set

$$A = \{x \in [a, b] : f(x) \leq u\}$$

Since $a \in A$, A is non-empty. Let $s = \sup(A)$. Suppose that $f(s) < u$. Since f is continuous, we can choose $\varepsilon = u - f(s)$. Then for some $\delta > 0$, $|f(x) - f(s)| < \varepsilon$ as long as $|x - s| < \delta$. In particular, consider $x = s + \frac{\delta}{2}$. Then we have

$$f\left(s + \frac{\delta}{2}\right) < f(s) + \varepsilon = u$$

Thus $s + \frac{\delta}{2} \in A$, a contradiction.

Now suppose that $f(s) > u$. Since f is continuous, we can choose $\varepsilon = f(s) - u$. Then for some $\delta > 0$, all x that satisfies $|x - s| < \delta$ also satisfy $|f(x) - f(s)| < \varepsilon$. In particular, consider $x = s - \frac{\delta}{2}$. Then we have

$$f\left(s - \frac{\delta}{2}\right) > f(s) - \varepsilon = u$$

Hence $s - \frac{\delta}{2}$ is an upper bound smaller than s, a contradiction. ■

Corollary 3.2.9
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a continuous function. Then $f(I)$ is an interval.

Proof
Suppose that $a, b \in I$. Then $f(a), f(b) \in f(I)$ and by the IVT, for every y between $f(a)$ and $f(b)$, you can find a corresponding x such that $f(x) = y$. ■

Theorem 3.2.10 (Boundedness of Continuous Functions)
Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f is bounded.

Proof
Suppose that f is unbounded. For each n, choose $x_n \in [a, b]$ such that $|f(x_n)| \geq n$. Since f is continuous, we can choose a subsequence (x_{n_k}) such that it converges to x. Since the interval is closed we must have $x \in [a, b]$. Then $f(x_{n_k}) \to f(x)$. But this is not possible since $f(x_{n_k})$ is becoming arbitrarily large. ■

Theorem 3.2.11 (Extreme Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f has a maximum and a minimum attained in the interval.

Proof
Let M be the supremum of $\{f(x) : x \in [a,b]\}$. Suppose that no point in the interval such that M is attained. Then the function $g(x) = M - f(x)$ is strictly positive and continuous on the interval. By the algebra of continuous functions, $\frac{1}{M-f(x)}$ is continuous and therefore bounded by, say R. Then $\frac{1}{R} \leq M - f(x)$ and hence $f(x) \leq M - \frac{1}{R}$. This shows that M is not the supremum, a contradiction. ■

3.3 Uniform Continuity

Definition 3.3.1 (Uniform Continuity)
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a function. Then f is uniformly continuous on I if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in I$ and

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

Example 3.3.2
Let $a, b > 0$. Let $f : [0, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \sqrt{ax^2 + b}$$

Then f is uniformly continuous.

Proof
Let $\varepsilon > 0$. Choose $\delta = \frac{\varepsilon}{\sqrt{a}}$. Let $x, y \in [0, \infty)$ such that $|y - x| < \delta$. Then we have

$$\begin{aligned}
\left|\sqrt{ay^2+b} - \sqrt{ax^2+b}\right| &= \left|\frac{(ay^2+b)-(ax^2+b)}{\sqrt{ay^2+b}+\sqrt{ax^2+b}}\right| \\
&= \left|\frac{a(y-x)(y+x)}{\sqrt{ay^2+b}+\sqrt{ax^2+b}}\right| \\
&\leq a\frac{|y+x|}{\sqrt{a}(y+x)}|y-x| \\
&= \sqrt{a}|y-x| \\
&= \sqrt{a}\delta \\
&= \varepsilon
\end{aligned}$$

so we are done. ■

Proposition 3.3.3
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a function. If f is uniformly continuous, then f is continuous.

Proof
Fix y and allow x to vary. Then we have the definition of continuity at $(y, f(y))$. ■

Proposition 3.3.4
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a continuous function. If I is a closed interval, then f is uniformly continuous.

Proof
Suppose that f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for all $\delta > 0$ there are points $x, y \in [a,b]$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| \geq \varepsilon$$

Now for every $k \in \mathbb{N}$, choose $x_k, y_k \in [a,b]$ such that $|x_k - y_k| < \frac{1}{k}$ and

$$|f(x_k) - f(y_k)| \geq \varepsilon$$

We have that (x_k) is bounded by $[a,b]$, thus by the Bolzano-Weierstrass theorem, there exists a convergent subsequence (x_{k_i}) such that it has a limit $x_0 \in [a,b]$. Mirror for y_k. Now we have

$$\begin{aligned}|x_0 - y_{k_i}| &\leq |x_0 - x_{k_i}| + |x_{k_1} - y_{k_i}| \\ &\leq |x_0 - x_{k_j}| + \frac{1}{k_j}\end{aligned}$$

Thus we also have $y_{k_j} \to x_0$. But since f is continuous at x_0, we must have

$$\left|f(x_{k_j}) - f(x_0)\right| < \frac{\varepsilon}{2}$$

and

$$\left|f(y_{k_j}) - f(x_0)\right| < \frac{\varepsilon}{2}$$

Thus

$$\begin{aligned}\left|f(x_{k_j}) - f(y_{k_j})\right| &\leq \left|f(x_{k_j}) - f(x_0)\right| + \left|f(y_{k_j}) - f(x_0)\right| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon\end{aligned}$$

This is a contradiction since we assumed that $|f(x) - f(y)| \geq \varepsilon$. ■

Proposition 3.3.5
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a uniformly continuous function. Suppose that $(s_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in I. Then $(f(s_n))_{n\in\mathbb{N}}$ is a Cauchy sequence.

Proof
Since s_n is Cauchy in I, then fix $\delta > 0$. There exists N such that $|s_n - s_m| < \delta$ for all $m, n > N$. Since f is uniformly continuous, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|s_n - s_m| < \delta \implies |f(s_n) - f(s_m)| < \varepsilon$$

for all $m, n > N$. Thus $f(s_n)$ is Cauchy. ■

4 Differentiation

4.1 Properties of the Derivative

Definition 4.1.1 (Derivative)
Let $f:(a,b)\to\mathbb{R}$ be a function and $c\in(a,b)$. We say f is differentiable at c or f has a derivative at c if the limit

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}$$

exists and is finite. We write $f'(c)$ or $\frac{df}{dx}|_{x=c}$ as the derivative of f at c. We say that f is differentiable if f is differentiable at all $c\in(a,b)$.

Example 4.1.2
The following are true.

- Let $k\in\mathbb{R}$. The function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=k$ is differentiable and $f'(x)=0$.
- The function $g:\mathbb{R}\to\mathbb{R}$ defined by $g(x)=x$ is differentiable and $f'(x)=1$.
- The function $h:[0,\infty)\to\mathbb{R}$ defined by $h(x)=\sqrt{x}$ is differentiable and $h'(x)=\frac{1}{2\sqrt{x}}$.

Proof
Let $c\in\mathbb{R}$. Then we have

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}=\lim_{x\to c}\frac{k-k}{x-c}=\lim_{x\to c}0=0$$

Hence f is differentiable and $f'(c)=0$.

Let $c\in\mathbb{R}$. Then we have

$$\lim_{x\to c}\frac{g(x)-g(c)}{x-c}=\lim_{x\to c}\frac{x-c}{x-c}=\lim_{x\to c}1=1$$

Hence g is differentiable and $g'(c)=1$.

Let $c\in[0,\infty)$. Then we have

$$\lim_{x\to c}\frac{h(x)-h(c)}{x-c}=\lim_{x\to c}\frac{\sqrt{x}-\sqrt{c}}{x-c}=\lim_{x\to c}\frac{x-c}{(x-c)(\sqrt{x}+\sqrt{c})}=\lim_{x\to c}\frac{1}{\sqrt{x}+\sqrt{c}}=\frac{1}{2\sqrt{c}}$$

Hence h is differentiable and $h'(c)=\frac{1}{2\sqrt{c}}$. ■

Lemma 4.1.3
Let $f:(a,b)\to\mathbb{R}$ be a function and $c\in(a,b)$. Then f is differentiable at c if and only if

$$\lim_{h\to 0}\frac{f(c+h)-f(c)}{h}$$

exists and is finite. In this case this limit is equal to $f'(c)$.

Proof
Simple change of variables where $x=c+h$. ■

Proposition 4.1.4
Let $I\subseteq\mathbb{R}$ be an interval. Let $f:I\to\mathbb{R}$ be a function. Let $c\in I$. If f is differentiable at c, then f is continuous at c.

Proof
Suppose that f is differentiable at c.

$$\begin{aligned}\lim_{x\to c} f(x)-f(c) &= \lim_{x\to c}\frac{(f(x)-f(c))(x-c)}{x-c}\\ &= f'(c)\lim_{x\to c}(x-c)\\ &= 0\end{aligned}$$

∎

Theorem 4.1.5 (Algebra of Derivatives)
Let $I\subseteq\mathbb{R}$ be an interval. Let $f,g:I\to\mathbb{R}$. Suppose that f and g are differentiable at $c\in I$. Then the following are true.

- Sum rule: $f+g$ is differentiable at c and $(f+g)'(c)=f'(c)+g'(c)$.
- Product rule: fg is differentiable at c and $(fg)'(c)=f'(c)g(c)+f(c)g'(c)$.
- Quotient rule: If $g'(c)\neq 0$, then f/g is differentiable at c and $(f/g)'(c)=\frac{f'(c)g(c)-f(c)g'(c)}{(g'(c))^2}$

Proof
Use the limit definition of the derivatives. ∎

Theorem 4.1.6 (Chain Rule)
Suppose that $g:(a,b)\to\mathbb{R}$ and $f:g((a,b))\to\mathbb{R}$ such that g is differentiable at $c\in(a,b)$ and f is differentiable at $g(c)\in g((a,b))$. Then $f\circ g$ is differentiable at c and $(f\circ g)'(c)=f'(g(c))g'(c)$.

Proof

$$\begin{aligned}\lim_{x\to c}\frac{f(g(x))-f(g(c))}{x-c} &= \lim_{x\to c}\frac{f(g(x))-f(g(c))}{g(x)-g(c)}\frac{g(x)-g(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(g(x))-f(g(c))}{g(x)-g(c)}\lim_{x\to c}\frac{g(x)-g(c)}{x-c}\\ &= f'(g(c))g'(c)\end{aligned}$$

∎

Definition 4.1.7 ($\mathcal{C}^n$-Functions)
Let $I\subseteq\mathbb{R}$ be an interval. Let $f:I\subseteq\mathbb{R}\to\mathbb{R}$ be a function. We say that f is a function of class $\mathcal{C}^n(I)$ if the following are true.

- $f^{(k)}:I\to\mathbb{R}$ is differentiable for $0\leq k\leq n-1$
- $f^{(n)}:I\to\mathbb{R}$ is continuous.

We say that f is a smooth function if f is $\mathcal{C}^n$ for all $n\in\mathbb{N}$.

Definition 4.1.8 (Smooth Functions)
Let $I\subseteq\mathbb{R}$ be an interval. Let $f:I\subseteq\mathbb{R}\to\mathbb{R}$ be a function. We say that f is a smooth function if $f\in\mathcal{C}^n(I)$ for all $n\in\mathbb{N}$. . In this case we write $f\in\mathcal{C}^\infty(I)$.

4.2 Inverse of the Derivative

Proposition 4.2.1
Let $I\subseteq\mathbb{R}$ be an interval. Let $f:I\to\mathbb{R}$ be a continuous and bijective function. Then $f^{-1}:f(I)\to I$ is continuous.

Proposition 4.2.2
Let $I\subseteq\mathbb{R}$ be an interval. Let $f:I\to\mathbb{R}$ be a continuous and bijective function. Suppose that f is

differentiable at $c \in I$ and $f'(c) \neq 0$. Then f^{-1} is differentiable at $f(c)$ and

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}$$

Theorem 4.2.3 (Inverse Function Theorem)
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a $\mathcal{C}^1$ function. Suppose that $f'(c) \neq 0$ for some $c \in I$. Then there exists an open interval $J \subseteq I$ such that $c \in J$ and the following are true.

- $f|_J$ is bijective.
- $f|_J^{-1}$ is a function of class $\mathcal{C}^1$.

4.3 Mean Value Theorem

Definition 4.3.1 (Local Maximum and Minimums)
Let $f : (a, b) \to \mathbb{R}$ be a function and $c \in (a, b)$.

- f is said to have a local maximum at c if $f(x) \leq f(c)$ for all $x \in (a, b)$
- f is said to have a local minimum at c if $f(x) \geq f(c)$ for all $x \in (a, b)$

Both local maximums and local minimums are called local extremums.

Theorem 4.3.2
If f is defined on an open interval containing c, if f assumes its maximum or minimum at c, and if f is differentiable at c, then

$$f'(c) = 0$$

Proof
Suppose that f attains maximum at c. If $x > c$, then $\frac{f(x)-f(c)}{x-c} \leq 0$ Thus

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \leq 0$$

If $x < c$, then $\frac{f(x)-f(c)}{x-c} \geq 0$. Thus

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \geq 0$$

Thus $f'(c) = 0$. The proof for minimum is similar. ■

Theorem 4.3.3 (Rolle's Theorem)
Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on the closed interval $[a, b]$ and differentiable at the open interval (a, b) and that $f(a) = f(b)$. Then there is a point c in the open interval where

$$f'(c) = 0$$

Proof
If f is constant, then its derivative is 0 everywhere. If f is non-constant, then by the theorem 4.2.9, f is bounded and by theorem 4.2.10, f attains a maximum and minimum in the interval. Those two points have their derivatives equal to 0. ■

Theorem 4.3.4 (The Mean Value Theorem)
Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). Then there is a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof

Consider the function $g(x) = f(x) - x\frac{f(b)-f(a)}{b-a}$. We have that

$$\begin{aligned} g(b) - g(a) &= f(b) - f(a) - (b-a)\frac{f(b)-f(a)}{b-a} \\ g(b) - g(a) &= 0 \\ g(b) &= g(a) \end{aligned}$$

Thus by Rolle's Theorem there exists a point c such that $g'(c) = 0$.

$$\begin{aligned} g'(c) &= 0 \\ f'(c) - \frac{f(b)-f(a)}{b-a} &= 0 \\ f'(c) &= \frac{f(b)-f(a)}{b-a} \end{aligned}$$

■

Theorem 4.3.5 (Generalized Mean Value Theorem)
Let f and g be continuous functions on $[a,b]$ that are differentiable on (a,b) Then there exists at least one x in (a,b) such that

$$f'(x)[g(b) - g(a)] = g'(x)[f(b) - f(a)]$$

Proof
Consider $h(x) = f(x)(g(b) - g(a)) - (f(b) - f(a))g(x)$. We have that $h(a) = h(b) = f(a)g(b) - g(a)f(b)$. By Rolle's Theorem there is a point between a and b where $h'(c) = 0$. Thus we have

$$\begin{aligned} h'(c) &= 0 \\ f'(c)(g(b) - g(a)) - g'(c)(f(b) - f(a)) &= 0 \end{aligned}$$

■

Theorem 4.3.6 (Constant Function)
If $f : (a,b) \to \mathbb{R}$ is differentiable and $f'(x) = 0$ for all $x \in (a,b)$ then f is constant.

Proof
Suppose that $h, k \in (a,b)$. By the Mean Value Theorem, there exists $c \in (h,k)$ such that

$$f'(c) = \frac{f(k) - f(h)}{k - h}$$

Since $f'(c) = 0$ we have that $f(h) = f(k)$ Since this is true for every $h, k \in (a,b)$, we have that

$$f(x) = m$$

for some $m \in \mathbb{R}$ for all $x \in (a,b)$. ■

Proposition 4.3.7
Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \to \mathbb{R}$ be a differentiable function. Then the following are true.

- If $\forall x \in I$, $f'(x) > 0$ then f is strictly increasing on the interval
- If $\forall x \in I$, $f'(x) < 0$ then f is strictly decreasing on the interval
- If $\forall x \in I$, $f'(x) \leq 0$ then f is increasing on the interval
- If $\forall x \in I$, $f'(x) \geq 0$ then f is decreasing on the interval

Proof
We demonstrate the proof only of the first item. $f'(x) > 0$ implies $\lim_{x\to c} \frac{f(x)-f(c)}{x-c} > 0$. If $x > c$, we must have $f(x) > f(c)$. If $x < c$, we must have $f(x) < f(c)$. Thus $f(x)$ is strictly increasing. ■

Theorem 4.3.8 (L'Hopital's Rule)
Let $I \subseteq \mathbb{R}$ be an interval. Let $c \in \mathbb{R}$. Let $f, g : I \setminus \{c\} \to \mathbb{R}$ be differentiable functions such that $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in I \setminus \{c\}$. Suppose that one of the following are true.

- $\lim_{x \to c} f(x) = 0$ and $\lim_{x \to c} g(x) = 0$.
- $\lim_{x \to c} g(x) = \pm\infty$.

Then we have
$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}$$

5 Integration

Students often misunderstand the importance of integration. It originated not from begin an inverse operation of differentiation, but as a tool to find the area under the function. It just so happens that they two serve as an inverse to each other. This point will be made very clear once you encounter differentiation and integration of multiple dimensions.

Integration has two underlying theories, that can be shown equivalent. These are the Darboux Integral and the Riemann Integral. We start our rigorous study on integration with the Darboux Integral.

5.1 Darboux Integral

The area under the function will be approximated by small rectangles, their height will be exactly the function while the collection of their various widths, will be called a partition. Once we establish most of the proporties of paritions we will then take limits.

Definition 5.1.1 (Partition)
Let $I = [a, b] \subseteq \mathbb{R}$ be a closed interval. A partition P of I is a set of the form

$$P = \{[t_0, t_1], [t_1, t_2], \dots, [t_{n-1}, t_n]\}$$

for $a = t_0 < t_1 < \cdots < t_n = b$.

A partition does not have to partition the x-axis equally. This means that rectangles can have different lengths.

For the heights, we obtain two approximations for the area, one that uses the shortest rectangle, and the other that uses the longest rectangles.

Definition 5.1.2 (Lower and Upper Sum)
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. Let $P = \{[t_0, t_1], [t_1, t_2], \dots, [t_{n-1}, t_n]\}$ be a partition of I.

- Let $m_k = \inf\{f(x) \mid x \in [t_k, t_{k+1}]\}$ for each $[t_k, t_{k+1}] \in P$. Define the lower sum of f to be

$$L(f, P) = \sum_{k=0}^{n-1} m_k(t_{k+1} - t_k)$$

- Let $M_k = \sup\{f(x) \mid x \in [t_k, t_{k+1}]\}$ for each $[t_k, t_{k+1}] \in P$. Define the lower sum of f to be

$$U(f, P) = \sum_{k=0}^{n-1} M_k(t_{k+1} - t_k)$$

As indicated by the notation, the lower and upper sum not only depends on the function, but it also depends on the partition one takes since the supremum and infinum of each particular rectangle would be different if two different partitions are chosen.

We now give an obvious proposition.

Proposition 5.1.3
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. Let P be a partition of I. Then we have

$$L(f, P) \le U(f, P)$$

Proof

For all $k \in \{1, \dots, n\}$, we have $m_k \leq M_k$. Thus

$$\sum_{k=1}^{n} m_k(t_k - t_{k-1}) \leq \sum_{k=1}^{n} M_k(t_k - t_{k-1})$$

and we are done. ∎

The above proposition states universally that the lower sum must be less than the upper sum. We will later see that the choice of parition need not matter, the lower sum will always be less than the upper sum. To formally prove that, we need the notion of a "better" partition.

Definition 5.1.4 (Refinement)
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. Let P, Q be partitions of I. We say that Q is a refinement of P if $P \subseteq Q$.

Be careful with the notion of refinement. It requires the intervals in P to be unions of intervals of Q. This means that there can be two partitions that are not refinements of one and another. One way to think of refinements easier is that every point in P must be contained in Q in order for intervals in P to be union of intervals in Q, which is why we use a subset notation to indicate refinements.

The above proposition is also obvious. As we make better approximations, the lower and upper sums get closer, which as you can guess, once the lower sum and upper sum are equal, that number will be the area under the function.

Proposition 5.1.5
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. Let P, Q be partitions of I such that Q is a refinement of P. Then we have

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$$

Proof
Let $P = \{t_0, \dots, t_n\}$. Let Q be a refinement of P. Then there exists some interval in P, say $[t_{i-1}, t_i]$ such that it is equal to $[s_{j-1}, s_j] \cup [s_j, s_{j+1}]$ for some j. Then $m_i \leq m_j, m_{j+1}$. Thus $m_i(t_i - t_{i-1}) \leq m_j(t_i - t_{i-1}), m_{j+1}(t_i - t_{i-1})$. Summing all these m_i and m_j we have that $L(f, P) \leq L(f, Q)$. The proof is mirrored for $U(f, Q) \leq U(f, P)$ ∎

We will need a lemma in order to prove the required result.

Lemma 5.1.6
Let I be a closed interval. Let P_1, P_2 be partitions of I. Then there exists a partition P of I such that P is a refinement of P_1 and P_2.

Proof
Let $P_1 = \{t_0, \dots, t_m\}$ and $P_2 = \{s_0, \dots, s_n\}$. Then define P by ordering all the elements of P_1 and P_2 from smallest to largest. Then $P_1 \subseteq P$ and $P_2 \subseteq P$. ∎

Below is the proposition we need to prove a lot of things. The trick is to make the common refinement so that comparison can be made between them.

Proposition 5.1.7
Let I be a closed interval. Let P_1, P_2 be partitions of I. Then we have

$$L(f, P_1) \leq U(f, P_2)$$

Proof
Let Q be the common refinement of P_1 and P_2. Then

$$L(f, P_1) \leq L(f, Q) \leq U(f, Q) \leq U(f, P_2)$$

■

Definition 5.1.8 (Lower and Upper Sum)
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function.

- Define the lower sum of f to be

$$\underline{\int_I} f = \sup\{L(f, P) \mid P \text{ is a partition of } I\}$$

- Define the upper sum of f to be

$$\overline{\int_I} f = \inf\{U(f, P) \mid P \text{ is a partition of } I\}$$

Definition 5.1.9 (Darboux Integrable Functions)
Let $I = [a, b]$ be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. We say that f is Darboux integrable if

$$\underline{\int_I} f = \overline{\int_I} f$$

In this case, we write the common number

$$\int_I f = \int_a^b f(x)\,dx$$

Proposition 5.1.10
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. Then f is integrable if and only if for every $\varepsilon > 0$ there exists a partition P of I such that $U(f, P) - L(f, P) < \varepsilon$.

Proof
Suppose that f is integrable. Let $T = \sup_P\{L(f, P)\} = \inf_P\{U(f, P)\}$. By the approximation property, fix $\frac{\varepsilon}{2} > 0$. There exists P_1, P_2 such that

$$T - \frac{\varepsilon}{2} < L(f, P_1) \leq T$$

and

$$T \leq U(f, P_2) < T + \frac{\varepsilon}{2}$$

Define a new partition P_3 such that $P_1 \subseteq P_3$ and $P_2 \subseteq P_3$, then

$$T - \frac{\varepsilon}{2} < L(f, P_1) \leq L(f, P_3) \leq T$$

and

$$T \leq U(f, P_3) \leq U(f, P_2) < T + \frac{\varepsilon}{2}$$

Then we have $T - L(f, P_3) < \frac{\varepsilon}{2}$ and $U(f, P_3) - T < \frac{\varepsilon}{2}$ and

$$U(f, P_3) - L(f, P_3) < \varepsilon$$

Now suppose that for every $\varepsilon > 0$, there exists P such that $U(f, P) - L(f, P) < \varepsilon$. Fix $\varepsilon > 0$. Then $U(f, P) - L(f, P) < \varepsilon$ implies $\inf_P\{U(f, P)\} - \sup_P\{L(f, P)\} < \varepsilon$. Thus for all $\varepsilon > 0$, $\inf_P\{U(f, P)\} - \sup_P\{L(f, P)\} < \varepsilon$ Thus we must have $\inf_P\{U(f, P)\} - \sup_P\{L(f, P)\}$ less than every positive number and larger than every negative number. Thus $\sup_P\{L(f, P)\} = \inf_P\{U(f, P)\}$. ■

This epsilon definition makes it easier to prove things since we are very familiar with closeness and limits. Often the supremum and infinum will be hard to compute which is why we will often resort to this theorem.

To end this section, we will discover another useful characterization of integrability. We start with a new definition on partitions.

Definition 5.1.11 (Mesh)
Let I be a closed interval. Let P be a partition of I. Define the mesh of P to be

$$\text{mesh}(P) = \max\{b - a \mid [a, b] \in P\}$$

Now we have the following equivalent characterization oif Darboux Integrability.

Proposition 5.1.12
Let I be a closed interval. Let $f : I \to \mathbb{R}$ be a bounded function. Then f is integrable if and only if for each $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\text{mesh}(P) < \delta \implies U(f, P) - L(f, P) < \varepsilon$$

for all partitions P of I.

Proof
The converse is trivial. Since there exists a $\delta > 0$ such that the condition hold, choose one of such P will complete the proof using the epsilon delta definition of Darboux Integrability.

Suppose that f is Darboux Integrable. Let $\varepsilon > 0$, select a partition P_0 such that

$$U(f, P_0) - L(f, P_0) < \frac{\varepsilon}{2}$$

Since f is bounded, there exists $B > 0$ such that $|f(x)| < B$ for all x. Let $\delta = \frac{\varepsilon}{8mB}$, where m is the number of partitions of P_0. Consider any partition P. We want to prove that $U(f, P) - L(f, P) < \varepsilon$. Let Q be the common refinement of P and P_0. I claim that

$$L(f, Q) - L(f, P) \leq 2mB \cdot \text{mesh}(P) < 2mB\delta = \frac{\varepsilon}{4}$$

When $m = 1$, Suppose that the refinement of P by P_0 is located at $[t_{k-1}, t_k]$. Then

$$\begin{aligned} L(f, Q) - L(f, P) &= \inf\{f(t) : t_{k-1} \leq t \leq u\}(u - t_{k-1}) + \inf\{f(t) : u \leq t \leq t_k\}(t_k - u) \\ &\quad - m_k(t_k - t_{k-1}) \\ &\leq B \cdot \text{mesh}(P) - (-B) \cdot \text{mesh}(P) && (-B \text{ is a minimum of } f(x)) \\ &< 2B\text{mesh}(P) \end{aligned}$$

For the case that Q has the maximum m elements not in P,

$$\begin{aligned} L(f, Q) - L(f, P) &\leq 2mB \cdot \text{mesh}(P) && (\text{Same argument as the case for } m = 1) \\ &< 2mB\delta \\ &= \frac{\varepsilon}{4} && (\text{constructed } \delta) \end{aligned}$$

Since Q is a refinement of P_0, we have $L(f, P_0) \leq L(f, Q)$ thus

$$L(f, P_0) - L(f, P) < \frac{\varepsilon}{4}$$

By a similar argument,

$$U(f, P) - U(f, P_0) < \frac{\varepsilon}{4}$$

Thus

$$U(f, P) - L(f, P) < U(f, P_0) - L(f, P_0) + \frac{\varepsilon}{2}$$

From the start of the proof, we have $U(f, P_0) - L(f, P_0) < \frac{\varepsilon}{2}$, thus we now have $U(f, P) - L(f, P) < \varepsilon$ as desired. ■

This gives some more restriction on the partition with a delta as well, and allows us to choose our partition better. These different notions of integrability will prove to be useful in different scenarious. Often in proving fundamental functions that cannot be decomposed with algebra of integrability, we will resort to these definitions. This is why it also crucial to learn how to properly find the required partitions for the epsilon, and epsilon-delta definitions.

5.2 Riemann Integral

The Riemann integral is more abstract thus standard lecture notes in developing the notion of integrability will most likely use the Darboux integral. Treat this section as a bonus.

Definition 5.2.1 (Tagged Partition)
A tagged partition is a partition $P = \{t_0, \dots, t_n\}$ such that every $[t_{k-1}, t_k]$ is associated with a $x_k \in [t_{k-1}, t_k]$.

Definition 5.2.2 (Riemann Sum)
Let f be a bounded function on $[a, b]$ and let $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$. A Riemann sum of f associated with the tagged partition P is a sum of the form

$$S(f, P) = \sum_{k=1}^{n} f(x_k)(t_k - t_{k-1}) = \sum_{k=1}^{n} x_k |I_k|$$

Definition 5.2.3 (Riemann Integrable Functions)
The function f is Riemann Integrable on $[a, b]$ if there exists a number L such that for every $\varepsilon > 0$, there exists δ such that $\text{mesh}(P) < \delta$ implies $|S(f, P) - L| < \varepsilon$ where P is a tagged partition.

As one can see, riemann integrability is more abstract in the sense that at least the Darboux integrability explicitly gives what the limit is. But here not only is the limit made implicitly, even the choice of the tag in the tagged partition is arbitrary.

Theorem 5.2.4
A bounded function f on $[a, b]$ is Riemann Integrable if and only if it is Darboux Integrable. In this case, both methods produce the same sum.

Proof
Suppose that f is Darboux Integrable. Then for any tagged partition P',

$$L(f, P) \leq S(f, P') \leq U(f, P)$$

Since it is Darboux Integrable,

$$U(f, P) < L(f, P) + \varepsilon \leq \sup_P \{L(f, P)\} + \varepsilon = \int_a^b f + \varepsilon$$

and

$$L(f, P) > U(f, P) - \varepsilon \geq \inf_P \{U(f, P)\} - \varepsilon = \int_a^b f - \varepsilon$$

Thus we have

$$\left| S(f, P') - \int_a^b f \right| < \varepsilon$$

Suppose that f is a bounded function that is Riemann Integrable and that converges to L. Let P be any partition of $[a, b]$. Then for every $\varepsilon > 0$, there exists a δ such that $\text{mesh}(P) < \delta$ implies $|S(f, P) - L| < \varepsilon$. In particular, choose a partition P such that $\text{mesh}(P) < \delta$. For each interval in P, choose the tag x_k such that

$$f(x_k) < m_k + \varepsilon$$

Thus we have
$$S(f,P) \leq L(f,P) + \varepsilon(b-a)$$
and
$$|S(f,P) - L| < \varepsilon$$
by definition of Riemann Integrability. Thus
$$\begin{aligned}\sup_P\{L(f,P)\} &\geq L(f,P)\\ &\geq S(f,P) - \varepsilon(b-a)\\ &> L - \varepsilon - \varepsilon(b-a)\end{aligned}$$
Thus we have $L \leq \sup_P\{L(f,P)\}$. Similarly we have $L \geq \inf_P\{U(f,P)\}$. Thus we have
$$\sup_P\{L(f,P)\} = L = \inf_P\{U(f,P)\}$$

■

Example 5.2.5
Define
$$f(x) = \begin{cases}\frac{1}{q} & \text{if } x = \frac{p}{q} \text{ where } p, q \in \mathbb{N} \text{ and } \gcd(p,q) = 1\\ 0 & \text{otherwise}\end{cases}$$
for $x \in [0,1]$. Then f is integrable. This example show that non-continuous functions can be integrable.

Example 5.2.6
Define
$$f(x) = \begin{cases}\frac{1}{q} & \text{if } x = \frac{p}{q} \text{ where } p, q \in \mathbb{N} \text{ and } \gcd(p,q) = 1\\ 0 & \text{otherwise}\end{cases}$$
for $x \in [0,1]$. Also define $g(x) = 1 - f(x)$. Then $f(x) + g(x)$ is integrable and
$$\inf_P\{U(f+g,P)\} \leq \inf_P\{U(f,P)\} + \inf_P\{U(g,P)\}$$
and
$$\sup_P\{L(f+g,P)\} \geq \sup_P\{L(f,P)\} + \sup_P\{L(g,P)\}$$
This example shows the upper and lower riemann sums of functions are not necessarily linear.

5.3 Properties of the Riemann Integral

This section gives us instant knowledge on whether a function is integrable, once we also have integrability of the basic functions.

We start the section with two sufficient and powerful conditions for integrability.

Theorem 5.3.1
Every monotonic function f on $[a,b]$ is integrable.

Proof
Consider a uniform partition P. We have

$$\begin{aligned}U(f,P)-L(f,P)&=\sum_{k=1}^{n}M_k(t_k-t_{k-1})-\sum_{k=1}^{n}m_k(t_k-t_{k-1})\\&\leq\frac{b-a}{n}\sum_{k=1}^{n}\left(f\left(a+\frac{k}{n}(b-a)\right)-f\left(a+\frac{k-1}{n}(b-a)\right)\right)\\&=\frac{b-a}{n}(f(b)-f(a))\end{aligned}$$

Thus given ε we can choose n large enough such that $\frac{(b-a)(f(b)-f(a))}{n}<\varepsilon$.

Theorem 5.3.2
Every continuous function $f:[a,b]\to\mathbb{R}$ is integrable.

Proof
Suppose that f is continuous. Then f is uniformly continuous in $[a,b]$. Pick a partition P such that $\text{mesh}(P)<\delta$, where δ is chosen from an ε from uniform continuity. Since f is continuous is in a closed interval, f is bounded and a maximum and minimum is achieved. For any M_k and m_k,

$$M_k-m_k=f(x_k)-f(y_k)<\frac{\varepsilon}{b-a}$$

by uniform continuity, where $x_k,y_k\in[t_{k-1},t_k]$. Hence

$$\begin{aligned}U(f,P)-L(f,P)&=\sum_{k=1}^{n}M_k(t_k-t_{k-1})-\sum_{k=1}^{n}m_k(t_k-t_{k-1})\\&=\sum_{k=1}^{n}(M_k-m_k)(t_k-t_{k-1})\\&<\sum_{k=1}^{n}\frac{\varepsilon}{b-a}(t_k-t_{k-1})\\&=\varepsilon\end{aligned}$$

The above two theorems already provide a rich foundation of integrable functions. These include the typical trigonometric functions, exponential functions, polynomials and more. However do be careful that being integrable does not means that you can alawys find a nice expression for the result, or even an existence of an antiderivative, as we will see in other sections.

We also have the basic addition and scalar multiplication of integrable functions.

Theorem 5.3.3 (Algebra of Integrals)
Let f,g be integrable functions on $[a,b]$, and let $c\in\mathbb{R}$. Then

- cf is integrable and $\int_a^b cf=c\int_a^b f$
- $f+g$ is integrable and $\int_a^b(f+g)=\int_a^b f+\int_a^b g$

Proof
We first note that

$$\sup_P\{L(cf,P)\}=\sup_P\{c\cdot L(f,P)\}=c\sup_P\{L(f,P)\}$$

and

$$\inf_P\{(U(cf,P)\}=\inf_P\{c\cdot U(f,P)\}=c\inf_P\{U(f,P)\}$$

if $c>0$. The first equality is achieved since $\sum_{k=1}^{n}cm_k(t_k-t_{k-1})=c\sum_{k=1}^{n}m_k(t_k-t_{k-1})$. The second equality is just a property of the supremum and infinum. This results in $\sup_P\{L(cf,P)\} = \inf_P\{(U(cf,P)\}$ by integrability of f. Thus cf is integrable. From those equality we can also deduce that

$$\int_a^b cf = \inf\{(U(cf,P)\} = c\inf\{U(f,P)\} = c\int_a^b f$$

The negative version can be proven just by setting $c=-1$. Note that $\sup_P(-f) = -\inf_P(f)$ and $\inf_P(-f) = -\sup_P(f)$. Thus $U(-f,P) = -L(f,P)$ and $L(-f,P) = -U(f,P)$ and

$$\inf\{U(-f,P)\} = -\sup\{L(f,P)\} = -\inf\{U(f,P)\} = \sup\{L(-f,P)\}$$

and we are done.

For the sum rule, note that $\sup(f+g) \leq \sup(f)+\sup(g)$ and thus

$$U(f+g,P) \leq U(f,P) + U(g,P)$$

and similarly

$$L(f,P) + L(g,P) \leq L(f+g,P)$$

Also since $U(f,P) - L(f,P) < \frac{\varepsilon}{2}$ and $U(g,P) - L(g,P) < \frac{\varepsilon}{2}$ for some P, we have $U(f+g,P) - L(f+g,P) < \varepsilon$. Thus we have $f+g$ is integrable.

Now

$$L(f,P) + L(g,P) \leq L(f+g,P) \leq U(f+g,P) \leq U(f,P) + U(g,P)$$

thus $\int_a^b f+g = \int_a^b f + \int_a^b g$ ■

Similar to sequences, we also have inequalities between values of the integral and the value of the function.

Theorem 5.3.4 (Monotinicity of the Integral)
If f,g are intergrable on $[a,b]$ and $f(x) \leq g(x)$ for all $x \in [a,b]$. Then $\int_a^b f \leq \int_a^b g$.

Proof
We note that since $g-f \geq 0$, $U(g-f,P) > \inf\{U(g-f,P)\} > 0$ for any partition P. By the algebra of integrals, $g-f$ is integrable and $\inf\{U(g-f,P)\} = \int_a^b g-f \geq 0$ which implies

$$\int_a^b f \leq \int_a^b g$$

■

With the above theorem in play, we can give the integral version of intermediate value theorem. But before that, we need a proposition.

Proposition 5.3.5
Let $f:[a,b] \to \mathbb{R}$ be integrable. Let $m = \inf_{x\in[a,b]}\{f(x)\}$ and $M = \sup_{x\in[a,b]}\{f(x)\}$. Then

$$m(b-a) \leq \int_a^b f \leq M(b-a)$$

Proof
Let $g = \sup\{f\}$ and $h(x) = \inf\{f\}$ be constant functions. Then $h(x) \leq f(x) \leq g(x)$ for all $x \in [a,b]$. By the above theorem, we have our desired inequality. ■

This can be made clear with a graph, the proposition simply states that the smallest rectangle above the function and the largest rectangle under the function bounds the area of the integral. Now we can give the intermediate value theorem.

Theorem 5.3.6 (Intermediate Value Theorem for Integration)
Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then there exists $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f$$

Proof
By the above proposition, we have

$$m \leq \frac{1}{b-a}\int_a^b f \leq M$$

Since f is continuous in the bounded interval $[a, b]$, f attains its maximum and minimum. Thus by the IVT, f must attain $\frac{1}{b-a}\int_a^b f$ for some $c \in (a, b)$. ■

Theorem 5.3.7
If f is integrable on $[a, b]$ then $|f|$ is integrable on $[a, b]$ and

$$\left|\int_a^b f\right| \leq \int_a^b |f|$$

Proof
Since we have the inequality $\sup\{|f|\} - \inf\{|f|\} \leq \sup\{f\} - \inf\{f\}$, we have

$$U(|f|, P) - L(|f|, P) < U(f, P) - L(f, P) < \varepsilon$$

and thus $|f|$ is integrable. Now by monotinicity, $-|f| \leq f \leq |f|$ and thus

$$\left|\int_a^b f\right| \leq \int_a^b |f|$$

■

Theorem 5.3.8
Let f be a function defined on $[a, b]$. f is integrable on $[a, c]$ and $[c, b]$ with $c \in (a, b)$ if and only if f is integrable on $[a, b]$ and

$$\int_a^b f = \int_a^c f + \int_c^b f$$

Proof
First suppose that $f : [a, b] \to \mathbb{R}$ is integrable. Then for every $\varepsilon > 0$ there exists a partition P such that $U(f, P) - L(f, P) < \varepsilon$. Let P_c be the refinement of P that contains c. Let Q be the partition of $[a, c]$ induced by P_c and R be the partition at $[c, b]$. Then we have

$$U(f, P_c) = U(f, Q) + U(f, R)$$

and

$$L(f, P_c) = L(f, Q) + L(f, R)$$

which means

$$U(f, Q) - L(f, Q) + U(f, R) - L(f, R) = U(f, Q) - L(f, Q) < \varepsilon$$

Note that the first two terms combined on the left hand side are positive and the last two terms as well thus they each must be less than ε and thus is integrable.

Now let $f : [a, c] \to \mathbb{R}$ and $f : [c, b] \to \mathbb{R}$ be integrable. For every $\frac{\varepsilon}{2}$ there exists partitions Q and R such that

$$U(f, Q) - L(f, Q) < \frac{\varepsilon}{2}$$

and

$$U(f,R) - L(f,R) < \frac{\varepsilon}{2}$$

Take P to be the common refinement of Q and R. Then

$$U(f,Q) - L(f,Q) = U(f,Q) - L(f,Q) + U(f,R) - L(f,R) < \varepsilon$$

thus f is integrable on $[a,b]$.

Note that we have

$$\begin{aligned}\int_a^b f &\leq U(f,P)\\ &= U(f,Q) + U(f,R)\\ &\leq L(f,Q) + L(f,R) + \varepsilon\\ &\leq \int_a^c f + \int_c^b f + \varepsilon\end{aligned}$$

and

$$\begin{aligned}\int_a^b &\geq L(f,P)\\ &= L(f,Q) + L(f,R)\\ &\geq U(f,Q) + U(f,R) - \varepsilon\\ &\geq \int_a^c f + \int_c^b f - \varepsilon\end{aligned}$$

Thus we have

$$\left(\int_a^c f + \int_c^b f\right) - \varepsilon \leq \int_a^b f \leq \left(\int_a^c f + \int_c^b f\right) + \varepsilon$$

■

The following theorem shows that composing a Riemann integrable function with a continuous function gives another Riemann integrable function. Mind the differences of this theorem and the theorem stated for the substitution rule in the later chapters.

Theorem 5.3.9
Let $f : [a,b] \to \mathbb{R}$ be a Riemann Integrable function and $g : \mathbb{R} \to \mathbb{R}$ is a continuous function. Then $g \circ f$ is Riemann Integrable.

Proof
Since f is integrable, f is bounded, thus we only need to consided $g : [-M,M] \to \mathbb{R}$. On this bounded interval, g is uniformly continuous and bounded, say $|g(x)| \leq K$.

Let $\varepsilon > 0$, set $\varepsilon' = \frac{\varepsilon}{2(b-a)}$. By uniform continuity, there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|g(x) - g(y)| < \varepsilon'$. Choose $\nu = \frac{\delta}{4K}\varepsilon$, by integrability of f, there exists Q such that

$U(f,Q) - L(f,Q) < \nu$. Let $h = g \circ f$ for convenience. Now consider $U(h,Q) - L(h,Q)$

$$\begin{aligned}
U(h,Q) - L(h,Q) &= \sum_{k=1}^{n} (\sup_{I_k}(h) - \inf_{I_k}(h))|I_k| \\
&= \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) < \delta}} \left(\sup_{I_k}(h) - \inf_{I_k}(h)\right)|I_k| + \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} \left(\sup_{I_k}(h) - \inf_{I_k}(h)\right)|I_k| \\
&\leq \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) < \delta}} \varepsilon'|I_k| + \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} (\sup_{I_k}(h) - \inf_{I_k}(h))|I_k| \\
&\qquad\qquad \text{(By the } \varepsilon' \text{ from uniform continuity)} \\
&\leq \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) < \delta}} \varepsilon'|I_k| + \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} 2K|I_k| \qquad \text{(By boundedness of } g\text{)} \\
&\leq \varepsilon' \sum_{k=1}^{n} |I_k| + 2K \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} |I_k| \\
&= \varepsilon'(b-a) + 2K \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} |I_k|
\end{aligned}$$

Now as a side note, observe that

$$\begin{aligned}
\sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} |I_k| &= \frac{1}{\delta} \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} \delta|I_k| \\
&< \frac{1}{\delta} \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} (\sup_{I_k}(f) - \inf_{I_k}(f))|I_k| \\
&\leq \frac{1}{\delta} \sum_{k=1}^{n} (\sup_{I_k}(f) - \inf_{I_k}(f))|I_k| \\
&= \frac{1}{\delta} \sum_{k=1}^{n} (M_k - m_k)|I_k| \\
&= \frac{1}{\delta}(U(f,Q) - L(f,Q)) \\
&< \frac{\nu}{\delta}
\end{aligned}$$

Thus continuing from the main calculation,

$$\begin{aligned}
\varepsilon'(b-a) + 2K \sum_{\substack{\sup_{I_k}(f) \\ -\inf_{I_k}(f) \geq \delta}} |I_k| &< \varepsilon'(b-a) + \frac{2K\nu}{\delta} \\
&= \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon
\end{aligned}$$

Thus we are done. ■

Theorem 5.3.10
Let $f, g : [a,b] \to \mathbb{R}$ be Riemann Integrable functions. Then fg and $\frac{f}{g}$ are both Riemann Integrable, with the second one provided that $\frac{1}{g}$ is bounded.

Proof

Note that if f is integrable, then f^2 is integrable by the above theorem, and choosing $g(x) = x^2$. Now we have that

$$fg = \frac{1}{2}((f+g)^2 - f^2 - g^2)$$

by the algebra of integrable functions, fg is integrable. For the quotient rule, note that if g is bounded, then there exists $\varepsilon > 0$ such that $\varepsilon < |g|$. Now consider

$$h(x) = \begin{cases} \frac{1}{x} & \text{if } |x| > \varepsilon \\ \frac{x}{\varepsilon^2} & \text{if } |x| \leq \varepsilon \end{cases}$$

Notice that $h \circ g = \frac{1}{g}$. Since h is continuous and g is integrable, $\frac{1}{g}$ is integrable by the above theorem. ■

The comoposition rule, although lengthy in proof, is extremely powerful as once can see. We are unable to prove the product and quotient rule without the fundamental theorem of calculus and so we will begin with it on the next section.

5.4 Fundamental Theorem of Calculus

Theorem 5.4.1 (Fundamental Theorem of Calculus I)

Suppose that $f : [a,b] \to \mathbb{R}$ is an integrable function and define the function $F : [a,b] \to \mathbb{R}$ by

$$F(x) = \int_a^x f(t)\,dt$$

Then F is continuous at $[a,b]$. Also if f is continuous at $c \in [a,b]$ then $F'(c) = f(c)$.

Proof

Note that since f is integrable, f is bounded by say M. Then

$$F(x+h) - F(x) = \int_x^{x+h} f(t)\,dt$$

by linearity. Considering the horizontal length and vertical length, we have

$$\begin{aligned} |F(x+h) - F(x)| &= \left| \int_x^{x+h} f(t)\,dt \right| \\ &\leq M|h| \end{aligned}$$

Thus F is continuous by choosing $\delta = \varepsilon$. Now

$$\lim_{h \to 0} \frac{1}{h} \int_x^{x+h} f(t)\,dt - f(x) = \lim_{h \to 0} \frac{1}{h} \int_x^{x+h} (f(t) - f(x))\,dt$$

We need to show that this is equal to 0. Now since f is continuous, given $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$. Therefore, choose $y = x + h$ and x to be x. For $|x + h - x| = |h| < \delta$,

$$\begin{aligned} \left| \frac{1}{h} \int_x^{x+h} (f(t) - f(x))\,dt \right| &\leq \left| \frac{1}{h} \int_x^{x+h} |f(t) - f(x)|\,dt \right| && \text{(By 6.3.7)} \\ &\leq \left| \frac{1}{h} \int_x^{x+h} \varepsilon\,dt \right| && \text{(By Uniform Continuity)} \\ &= \varepsilon \end{aligned}$$

Thus we have

$$\lim_{h \to 0} \frac{1}{h} \int_x^{x+h} (f(t) - f(x))\,dt = 0$$

and

$$\lim_{h\to 0}\frac{1}{h}\int_x^{x+h} f(t)\,dt = f(x)$$

Thus $F'(x) = f(x)$. ■

Theorem 5.4.2 (Fundamental Theorem of Calculus II)
Let $F : [a, b] \to \mathbb{R}$ be a continuous function that is differentiable on (a, b) with $F' = f$. Suppose that $f : [a, b] \to \mathbb{R}$ is an integrable function. Then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Proof
Consider any partition P of the interval $[a, b]$. For every interval in P,

$$\inf_{I_k}(f(x))(t_k - t_{k-1}) \leq f(c_k)(t_k - t_{k-1}) \leq \sup_{I_k}(f(x))(t_k - t_{k-1})$$

for every $c_k \in (t_{k-1}, t_k)$. Since F is continuous on $[t_{k-1}, t_k]$ and differentiable on (t_{k-1}, t_k), by the mean value theorem, there exists c_k such that $F(t_k) - F(t_{k-1}) = f(c_k)(t_k - t_{k-1})$. Thus

$$\inf_{I_k}(f(x))(t_k - t_{k-1}) \leq F(t_k) - F(t_{k-1}) \leq \sup_{I_k}(f(x))(t_k - t_{k-1})$$

and

$$L(f, P) \leq \sum_{k=1}^{n}(F(t_k) - F(t_{k-1}) \leq U(f, P)$$

and

$$L(f, P) \leq F(b) - F(a) \leq U(f, P)$$

This is true for every P thus

$$\sup_P(L(f, P)) \leq F(b) - F(a) \leq \inf_P(U(f, P))$$

and thus

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

■

For the first time, we are given an explicit way of calculating the integral, given that the antiderivative exists. This is another reason why the fundamental theorem of calculus is so important. Without it, we will need to resort to the basic definitions of partitions.

Theorem 5.4.3
Let $f : [a, b] \to \mathbb{R}$ be an integrable function on $[a, b]$ and continuous at a. Let $x \in I_h$ and $|I_h| \to 0$. Then

$$\lim_{h\to 0}\frac{1}{|I_h|}\int_{I_h} f(t)\,dt = f(x)$$

Proof
From the last part of the proof of the FTC I, this is already proven. ■

Theorem 5.4.4 (Product Rule of Integrals)
Let $f, g : [a, b] \to \mathbb{R}$ be continuous functions on $[a, b]$ that are differentiable on (a, b) and such that f', g' are integrable on $[a, b]$. Then

$$\int_a^b f(x)g'(x)\,dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x)\,dx$$

Proof
Note that by the algebra of integrable function, $f'g, fg', (fg)'$ are all integrable. Also by the FTC, we have
$$\int_a^b (fg)' = f(b)g(b) - f(a)g(a)$$
and
$$\int_a^b (fg)' = \int_a^b f'g + fg'$$
thus we have the required result. ■

Theorem 5.4.5 (Chain Rule of Integrals)
Let $f : [a, b] \to \mathbb{R}$ be a differentiable function such that f' is integrable on $[a, b]$. Let g be a continuous function on $f([a, b])$. Then
$$\int_a^b g\left(f(x)\right) f'(x)\, dx = \int_{f(a)}^{f(b)} g(t)\, dt$$

Proof
Define $G(x) = \int_{f(a)}^x g(t)\, dt$. By the FTC, $G'(x) = g(x)$. By the chain rule of differentiation, we obtain $G'(f(x)) = g(f(x))f'(x)$. Since g is continuous and f is integrable, $g \circ f$ is integrable. Also f' is integrable. By the algebra of integrable functions, $G'(f(x))$ is integrable. Thus we have
$$\begin{aligned}\int_a^b g(f(x))f'(x)\, dx &= \int_a^b G'(f(x))\, dx \\ &= G(f(b)) - G(f(a)) && \text{(FTC)} \\ &= \int_{f(a)}^{f(b)} g(t)\, dt && \text{(FTC)}\end{aligned}$$
■

A neat trick to apply the FTC on non-standard integral bounds is to use the following formula:
$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(t)\, dt = b'(x)f(b(x)) - a'(x)f(a(x))$$
This involves using the chain rule. The proof is simple with the given conditions for both the FTC and the chain rule.

5.5 Improper Integrals

Definition 5.5.1 (Improper Integral)
Let $f : [a, b] \to \mathbb{R}$ be Riemann Integrable for every $[c, b]$ with $a < c$. Then the improper integral of f on $[a, b]$ is defined as
$$\int_a^b f(x)\, dx = \lim_{\varepsilon \to 0^+} \int_{a+\varepsilon}^b f(x)\, dx$$
Similarly, if $f : [a, b] \to \mathbb{R}$ is Riemann Integrable for every $[a, c]$ with $c < b$, then the improper integral of f on $[a, b]$ is define as
$$\int_a^b f(x)\, dx = \lim_{\varepsilon \to 0^+} \int_a^{b-\varepsilon} f(x)\, dx$$

Definition 5.5.2 (Improper Integrals on Interior Points)
Let $f : [a, b] \to \mathbb{R}$ be a function that is integrable on $[a, c - \varepsilon]$ and $[c + \delta, b]$ for all $\varepsilon, \delta > 0$ where the

interval makes sense. Then define the integral of f on $[a, b]$ to be

$$\int_a^b f(x)\,dx = \lim_{\varepsilon\to 0^+}\int_a^{c-\varepsilon} f(x)\,dx + \lim_{\delta\to 0^+}\int_{c+\delta}^b f(x)\,dx$$

Definition 5.5.3 (Improper Integrals with Infinity)
let $f : [a, \infty) \to \mathbb{R}$ be a function that is integrable for every interval $[a, c]$ with $a < c < \infty$. We define the improper integral of f on $[a, \infty]$ by

$$\int_a^\infty f(x)\,dx = \lim_{c\to\infty}\int_a^c f(x)\,dx$$

Similarly if $g : (-\infty, b] \to \mathbb{R}$ is an integrable function for every integral $[c, b]$ with $-\infty < c < b$ then we define the improper integral of g on $(-\infty, b]$ by

$$\int_{-\infty}^b g(x)\,dx = \lim_{c\to-\infty}\int_c^b g(x)\,dx$$

Finally, if $h : \mathbb{R} \to \mathbb{R}$ is a function that is integrable on every bounded interval $[a, b]$, then define the improper integral

$$\int_{-\infty}^\infty f(x)\,dx = \lim_{a\to-\infty}\int_a^c f(x)\,dx + \lim_{b\to\infty}\int_c^b f(x)\,dx$$

where c is any point in $\mathbb{R}$.

Theorem 5.5.4 (Absolute Comparison Test)
Let $f : [a, \infty) \to \mathbb{R}$ be integrable on $[a, b]$. For every $b > a$. If $\int_a^\infty |f| < \infty$, then $\int_a^\infty f$ converges. Moreover, if $g : [a, \infty) \to [0, \infty)$ is such that $|f| \leq g$ and $\int_a^\infty g < \infty$, then $\int_a^\infty f$ is absolutely convergent.

Proof
Recall from Cauchy Criterion that $\lim_{t\to\infty}\int_a^t |f|$ is finite if and only if for every $\varepsilon > 0$, there exists N such that $R_1, R_2 > N$ implies

$$\left|\int_a^{R_2} |f| - \int_a^{R_1} |f|\right| = \left|\int_{R_1}^{R_2} |f|\right| < \varepsilon$$

Thus we have that

$$\left|\int_{R_1}^{R_2} f\right| \leq \int_{R_1}^{R_2} |f| < \varepsilon$$

So $\int_a^\infty$ converges by the Cauchy Criterion.

Similarly, we have that for every $\varepsilon > 0$, there exists N such that $R_1, R_2 > N$ implies

$$\left|\int_a^{R_2} g - \int_a^{R_1} g\right| = \left|\int_{R_1}^{R_2} g\right| < \varepsilon$$

Therefore

$$\left|\int_{R_1}^{R_2} |f|\right| \leq \left|\int_{R_1}^{R_2} g\right| < \varepsilon$$

■

6 Series

6.1 Series and Convergences

Definition 6.1.1 (Definition of Series)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Define the series with nth term a_n to be the sequence $(s_n)_{n\in\mathbb{N}}$ given by

$$s_n = \sum_{i=0}^{n} a_i$$

We denote the limit of the sequence by $\sum_{k=0}^{\infty} a_k$.

Example 6.1.2 (Geometric Series)
The geometric series

$$\sum_{k=0}^{\infty} ar^k$$

where $a, r \in \mathbb{R}$ and $a \neq 0$, converges if and only if $|r| < 1$. In this case

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$$

Proof
Consider the partial sum $s_n = a + ar + \cdots + ar^n$. We have $s_n = \frac{1-r^{n+1}}{1-r}$. If $|r| < 1$, then (r^{n+1}) converges to 0. Then for every $(1-r)\varepsilon > 0$ there exists some N such that $n \geq N$ implies $\left|r^{n+1}\right| < (1-r)\varepsilon$. Then

$$\begin{aligned} \left|\frac{1-r^{n+1}}{1-r} - \frac{1}{1-r}\right| &= \left|\frac{r^{n+1}}{1-r}\right| \\ &< \varepsilon \end{aligned}$$

Thus the geometric series converges if and only if $|r| < 1$. ■

Proposition 6.1.3 (Null Sequence Test)
If $(a_n)_{n\in\mathbb{N}}$ does not tend to 0 then $\sum_{k=0}^{\infty} a_k$ diverges.

Proof
Suppose that $\sum_{n=0}^{\infty} a_n$ converges. Then we have that for all $\varepsilon > 0$, there exists some N such that

$$\left|\sum_{k=0}^{m} a_k - \sum_{k=0}^{n} a_k\right| < \varepsilon$$

whenever $n, m > N$. In particular, choosing $m = n+1$, we have

$$\begin{aligned} \left|\sum_{k=0}^{n+1} a_k - \sum_{k=0}^{n} a_k\right| &= |a_{n+1}| \\ &< \varepsilon \end{aligned}$$

for all $n > N$. Thus we have that $(a_n) \to 0$. ■

6.2 Series with Non-negative Terms

Proposition 6.2.1
If $a_n \geq 0$ for all $n > N$ with $N \in \mathbb{N}$ then the series $\sum_{k=1}^{\infty} a_k$ converges in $\mathbb{R}$ if and only if its partial sums are bounded.

Proof
We first prove the forward implication. Suppose that $\sum_{k=1}^{\infty} a_k$ converges, then by 2.1.4 it is bounded. Now suppose that $\sum_{k=1}^{\infty} a_k$ is bounded. Then since $a_n \geq 0$ then $\sum_{k=1}^{\infty} a_k$ is increasing. Then $\sum_{k=1}^{\infty} a_k$ is convergent by monotonic increasing theorem. ■

Theorem 6.2.2 (Direct Comparison Test)
Suppose $0 \leq a_n \leq b_n$ for all $n > N$ with $N \in \mathbb{N}$.
- If $\sum_{k=1}^{\infty} b_k$ converges then $\sum_{k=1}^{\infty} a_k$ converges.
- If $\sum_{k=1}^{\infty} a_k$ diverges then $\sum_{k=1}^{\infty} b_k$ diverges.

Proof
Suppose $0 \leq a_n \leq b_n$ for all $n > N$.

- Suppose that $\sum_{k=1}^{\infty} b_k$ converges to b. Then since $a_n \leq b_n$ for all n then a_n is bounded above by b. Then by the monotonic increasing theorem a_n converges.
- Suppose that $\sum_{k=1}^{\infty} a_k$ diverges. Then for every $C > 0$ there exists N such that $\sum_{k=1}^{\infty} a_k > C$ for all $n > N$. Then
$$C < \sum_{k=1}^{\infty} a_k \leq \sum_{k=1}^{\infty} b_k$$
thus we also have $\sum_{k=1}^{\infty} b_k \geq C$. Thus $\sum_{k=1}^{\infty} b_k$ diverges.

■

Example 6.2.3
The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges.

Proof
Let $s_n = \sum_{k=1}^{n} \frac{1}{k}$. Then
$$\begin{aligned} s_{2n} &= \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n-1} + \frac{1}{2n} + s_n \\ &\geq \frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n} + \frac{1}{2n} + s_n \\ &= s_n + \frac{1}{2} \end{aligned}$$
Now, we have
$$\begin{aligned} s_{2^n} &\geq s_{2^{n-1}} + \frac{1}{2} \\ &\geq s_{2^{n-2}} + \frac{1}{2} + \frac{1}{2} \\ &\geq s_1 + \frac{1}{2} + \cdots + \frac{1}{2} && (n \text{ times}) \\ &= 1 + \frac{n}{2} \end{aligned}$$
By the direct comparison test, we now have s_n diverge since $1 + \frac{n}{2}$ diverges. ■

Example 6.2.4
$\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges.

Proof
We first prove that $\sum_{k=2}^{\infty} \frac{1}{k^2-k}$ converges. Consider $s_n = \sum_{k=2}^{n} \frac{1}{(k)(k-1)}$.

$$\begin{aligned} s_n &= \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ &= \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1} \right) - \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1} + \frac{1}{n} \right) \\ &= 1 - \frac{1}{n} \end{aligned}$$

Thus we have that $s_n \to 1$. By the direct comparison test, we have that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges. ■

Theorem 6.2.5 (p series test)
Let $p \in \mathbb{R}$ be a real number. Then $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges when $p > 1$ and diverges when $0 < p \leq 1$.

Proof
When $0 < p \leq 1$, we have that $\sum_{k=1}^{\infty} \frac{1}{n^p}$ diverges by the direct comparison test with $\sum_{k=1}^{\infty} \frac{1}{n}$. Now suppose that $p > 1$. Let $S_n = \sum_{k=1}^{n} \frac{1}{k^p}$. Let $m \in \mathbb{N}$. Since $(S_n)_{n \in \mathbb{N}}$ is a strictly increasing sequence, we have

$$\begin{aligned} S_{2m+1} &= \sum_{k=1}^{2m+1} \frac{1}{k^p} \\ &= 1 + \sum_{i=1}^{m} \left(\frac{1}{(2i)^p} + \frac{1}{(2i+1)^p} \right) \\ &< 1 + \sum_{i=1}^{m} \frac{2}{(2i)^p} \\ &= 1 + \sum_{i=1}^{m} \frac{2^{1-p}}{i^p} \\ &= 1 + 2^{1-p} S_m \\ &< 1 + 2^{1-p} S_{2m+1} \end{aligned}$$

Rearranging gives $S_{2m+1} < \frac{1}{1-2^{1-p}}$. Again since the sequence is strictly increasing, we have that $S_n < \frac{1}{1-2^{1-p}}$ for all $n \in \mathbb{N}$. Since $(S_n)_{n \in \mathbb{N}}$ is increasing and bounded above, we conclude that it is convergent. ■

Theorem 6.2.6 (Limit Comparison Test)
Let $a_n, b_n > 0$ for all $n > N$ with $N \in \mathbb{N}$. If $\frac{a_n}{b_n}$ converges to $x \in (0, \infty)$, then $\sum_{k=1}^{\infty} b_k$ converge if and only if $\sum_{k=1}^{\infty} a_k$ converges.

Proof
We know that for every $\varepsilon > 0$ there exists N such that

$$\left| \frac{a_n}{b_n} - x \right| < \varepsilon$$

whenever $n > N$. This means that

$$(x - \varepsilon) b_n < a_n < (x + \varepsilon) b_n$$

As $x > 0$, choose ε so that $x - \varepsilon > 0$. Then $b_n < \frac{1}{x-\varepsilon} a_n$ thus by the direct comparison test if $\sum_{k=1}^{\infty} a_k$ converges then $\sum_{k=1}^{\infty} b_k$ converges. Similarly, $\frac{1}{c+\varepsilon} < b_n$ so if $\sum_{k=1}^{\infty} a_k$ diverges, then

$\sum_{k=1}^{\infty} b_k$ diverges. ■

Theorem 6.2.7 (Root Test)
Let $a_n \geq 0$ for all n and let $p = \lim_{n\to\infty} a_n^{1/n}$.

- If $p < 1$ then $\sum_{k=1}^{\infty} a_k$ converges
- If $p > 1$ then $\sum_{k=1}^{\infty} a_k$ diverges

Proof
We first proof the case with $p < 1$. Choose $\varepsilon > 0$ such that $p + \varepsilon < 1$. Then for all $n > N$, we have $a_n^{\frac{1}{n}} < (p+\varepsilon)$ and $a_n \leq (p+\varepsilon)^n$. Since $p + \varepsilon < 1$, $\sum_{k=0}^{\infty}(p+\varepsilon)^k$ converges and by comparison test, $\sum_{k=1}^{\infty} a_k$ converges.

Now suppose that $p > 1$ then choose $\varepsilon > 0$ such that $p - \varepsilon > 1$. Then for all $n > N$ we have $a_n^{\frac{1}{n}} > p + \varepsilon$ thus $a_n > (p+\varepsilon)^n$. Then $\sum_{n=0}^{\infty}(p+\varepsilon)^n$ diverges by the geometric series and $\sum_{k=1}^{\infty} a_k$ diverges by comparison test. ■

Theorem 6.2.8 (Ratio Test)
Let $a_n > 0$ for all $n > N$ with $N \in \mathbb{N}$. Let $r = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$

- If $r < 1$, then $\sum_{k=1}^{\infty} a_k$ converges.
- If $r > 1$, then $\sum_{k=1}^{\infty} a_k$ diverges.

Proof
Suppose that $r < 1$. Then choose

$$0 < \varepsilon = \frac{1-r}{2} < 1$$

Then there exists some N such that for every $n > N$,

$$\left|\frac{a_{n+1}}{a_n} - r\right| < \frac{1-r}{2} \implies \frac{a_{n+1}}{a_n} < \frac{1+r}{2} < 1$$

This means that $a_{n+1} < (\frac{1+r}{2})a_n$ and $a_{n+1} < (\frac{1+r}{2})^n a_1$. Since $\frac{1+r}{2} < 1$, $\sum_{n=0}^{\infty}(\frac{1+r}{2})^n$ converges by geometric series and $\sum a_n$ converges by comparison test.

Similarly for the divergence case, choose

$$\varepsilon = \frac{r-1}{2}$$

Then there exists some N such that for every $n > N$,

$$\left|\frac{a_{n+1}}{a_n} - r\right| < \frac{1-r}{2} \implies 1 < \frac{r+1}{2} < \frac{a_{n+1}}{a_n}$$

Then $a_n(\frac{r+1}{2}) < a_{n+1}$ and $a_1(\frac{r+1}{2})^n < a_{n+1}$. By geometric series $\sum_{n=0}^{\infty}(\frac{r+1}{2})^n$ diverges and by comparison test $\sum_{k=1}^{\infty} a_k$ diverges. ■

6.3 The Integral Test for Series

In this section here we suppose that the techniques and origins of integration is already properly introduced, it would be weird to add the integral test to the section of integration since it is developed for testing series convergence.

Theorem 6.3.1 (Integral Bounds)
Let f be decreasing and non-negative and integrable on the interval $[1, \infty)$. Then

$$\int_{m+1}^{n+1} f(x)\,dx \leq \sum_{k=m+1}^{n+1} f(k) \leq \int_{m}^{n} f(x)\,dx$$

Proof
The inequality is obtained through drawing the graph of a decreasing function f and estimating the area of the curve with rectangles of width 1.

Theorem 6.3.2 (Integral Test)
Let f be decreasing and non-negative and integrable on the interval $[1, \infty)$. Then $s_n = \sum_{k=1}^{n} f(k)$ converges if and only if $\int_1^\infty f(x)\,dx$ converges.

Proof
From the integral bounds, we have

$$\sum_{k=0}^{\infty} f(k) \leq \int_{1}^{\infty} f(x)\,dx$$

Thus when the integral is converges, the increasing sum is bounded. Thus by the monotone convergence theorem, the $\sum_{k=0}^{\infty}$ converges. When the integral diverges, then

$$\int_{0}^{\infty} f(x)\,dx \leq \sum_{k=0}^{\infty} f(k)$$

also diverges by the comparison test.

6.4 Euler's Number

Lemma 6.4.1
The infinite sum

$$\sum_{k=0}^{\infty} \frac{1}{k!}$$

converges.

Proof
For all $n > 0$, we have that $n! \geq 2^{n-1}$, we have that $\frac{1}{n!} \leq \frac{1}{2^{n-1}}$. Thus we have that

$$\sum_{k=0}^{n} \frac{1}{k!} \leq \sum_{k=0}^{n} \frac{1}{2^k}$$

$\sum_{k=0}^{n} \frac{1}{2^k}$ is a geometric series so it converges then it has an upper bound. By the completeness axiom, we have that the sum converges.

Definition 6.4.2 (Euler's Number)
Define Euler's number to be

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

Lemma 6.4.3
The series

$$a_n = \left(1 + \frac{1}{n}\right)^n$$

converges.

Proof
Consider $\frac{a_{n+1}}{a_n}$. We have that $\frac{a_{n+1}}{a_n} = \left(1+\frac{1}{n+1}\right)\left(1-\frac{1}{(n+1)^2}\right)^n$. Note that since $\frac{-1}{(n+1)^2} > -1$, we can apply the Bernoulli's Inequality can be applied. Then

$$\left(1+\frac{1}{n+1}\right)\left(1-\frac{1}{(n+1)^2}\right)^n \geq \left(1+\frac{1}{n+1}\left(1-\frac{n}{(n+1)^2}\right)\right) = 1+\frac{1}{(n+1)^3} \geq 1$$

Thus a_n is an increasing sequence.

Now note that $\left(1+\frac{1}{2n}\right)^n = \frac{1}{\left(1+\frac{1}{2n+1}\right)^n}$. Since $\frac{-1}{2n+1} > -1$. Thus we can apply Bernoulli's Inequality to have

$$\begin{aligned} \left(1-\frac{1}{2n+1}\right)^n &\geq 1-\frac{n}{2n+1} \\ \left(1-\frac{1}{2n+1}\right)^n &\geq \frac{n+1}{2n+1} \\ \frac{2n+1}{n+1} &\geq \frac{1}{\left(1-\frac{1}{2n+1}\right)^n} \\ 2-\frac{1}{n+1} &\geq \left(1+\frac{1}{2n}\right)^n \\ 2 &\geq \left(1+\frac{1}{2n}\right)^n \end{aligned}$$

Then we have $0 \leq a_{2n} \leq 4$, which is bounded. By the completeness axiom, this sequence converges. ■

Proposition 6.4.4

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{\infty}\frac{1}{k!}$$

Proof
We start by considering the nth term of both sequences. From the binomial theorem, we have that

$$\begin{aligned} \left(1+\frac{1}{n}\right)^n &= \sum_{k=0}^{n}\frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) \\ &\leq \sum_{k=0}^{n}\frac{1}{k!} \end{aligned}$$

Thus letting $n\to\infty$, we have

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \leq \sum_{n=0}^{\infty}\frac{1}{n!}$$

However, if $n > m$,

$$\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{m}\frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\cdots\left(1-\frac{k-1}{n}\right)\right)$$

Letting $n\to\infty$, we have $e \geq s_m$. Letting $m\to\infty$, we have $e \geq s$. Thus we have $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$ ■

Lemma 6.4.5
The number e is irrational.

Proof
Suppose that e can be written in an irreducible fraction $\frac{p}{q}$.

$$\begin{aligned} e-\sum_{k=1}^{q+1}\frac{1}{(k-1)!} &= \frac{p}{q}-\left(\frac{1}{0!}+\frac{1}{1!}+\dots+\frac{1}{q!}\right) \\ &= \frac{p(q-1)!-(q!+q!+\frac{q!}{2}+\dots+1)}{q!} \end{aligned}$$

Then $k=p(q-1)!-(q!+q!+\frac{q!}{2}+\dots+1)\in\mathbb{N}$ and $e-\sum_{k=1}^{q+1}\frac{1}{(k-1)!}=\frac{k}{q!}$. Now consider

$$\begin{aligned} e-\sum_{k=1}^{q+1}\frac{1}{(k-1)!} &= \sum_{k=0}^{\infty}\frac{1}{k!}-\sum_{k=0}^{q}\frac{1}{k!} \\ &= \frac{1}{(q+1)!}+\frac{1}{(q+2)!}+\dots \\ &= \frac{1}{q!}\left(\frac{1}{q+1}+\frac{1}{(q+1)(q+2)}+\dots\right) \\ &< \frac{1}{q!}\left(\frac{1}{2}+\frac{1}{2}\cdot\frac{1}{2}+\dots\right) && \text{(since } q>1\text{)} \\ &= \frac{1}{q!}\left(\frac{1}{2}+\frac{1}{4}+\dots\right) \\ &= \frac{1}{q!} && \text{(by geometric series)} \end{aligned}$$

Thus now we have that $\frac{k}{q!}<\frac{1}{q!}$ for some $k\in\mathbb{N}$ which is not possible for any choice of k. Thus we have arrived in a contradiction. ■

6.5 Alternating Series

Definition 6.5.1 (Alternating Series)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence such that $a_n\geq 0$ for all $n\in\mathbb{N}$. Define the alternating series of the sequence to be the series whose nth term is given by

$$s_n=\sum_{k=0}^{\infty}(-1)^k a_k$$

Theorem 6.5.2 (Alternating Series Test)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Suppose that the sequence is decreasing and converges to 0. Then $\sum_{k=0}^{\infty}(-1)^k a_k$ converges.

Proof
Firstly note that

$$s_{2n+2}=\sum_{k=1}^{2n}(-1)^k a_k-a_{2n+1}+a_{2n+2}$$

Since a_n is decreasing, $a_{2n+1}\geq a_{2n+1}$ thus $s_{2n+2}\geq s_{2n}$. Similarly, we have $s_{2n+1}\leq s_{2n-1}$. Combining the two inequality, we have

$$s_2\leq s_{2n}+a_{2n+1}=s_{2n+1}\leq s_1$$

Since s_{2n} is increasing and bounded, it converges. Similarly, s_{2n+1} is decreasing and bounded thus converges. Suppose that $(s_{2n}) \to s$. Then $s_{2n} + a_{2n+1} = s_{2n+1}$. Thus $(s_{2n+1}) \to s + 0 = s$. Suppose that for all ε, there exists N such that $|a_n| < \varepsilon$ whenever $n > N$. Then, we have

$$\begin{aligned} s_{2n} &\le s \\ s_{2n+1} - a_{2n+1} &\le s \\ s_{2n+1} - s &\le a_{2n+1} \\ |s_{2n+1} - s| &\le a_{2n+1} \\ |s_{2n+1} - s| &< \varepsilon \end{aligned}$$

Similarly,

$$\begin{aligned} s &\le s_{2n-1} \\ s &\le s_{2n} - a_{2n-1} \\ s - s_{2n} &\le a_{2n-1} \\ |s_{2n} - s| &\le a_{2n-1} \\ |s_{2n} - s| &< \varepsilon \end{aligned}$$

Thus we have that $|s_n - s| < \varepsilon$ whenever $n > N$ and (s_n) converges. ■

Example 6.5.3
The alternating harmonic series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

converges.

Proof
Simple application of the alternating series test. ■

6.6 Absolute Convergence

Definition 6.6.1 (Absolute Convergence)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. We say that the series with nth term a_n converges absolutely if the series $\sum_{k=0}^{\infty} |a_n|$ converges absolutely.

Proposition 6.6.2
Every absolutely convergent series is convergent.

Proof
Consider $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} |a_n|$. Without loss of generality suppose that $n > m$

$$\begin{aligned} |s_n - s_m| &= |a_n + a_{n-1} + \cdots + a_{m+1}| \\ &\le |a_n| + |a_{n-1}| + \cdots + |a_{m+1}| \\ &= t_n - t_m \\ &= |t_n - t_m| \end{aligned}$$

If (t_n) converges then for every $\varepsilon > 0$ there exists N such that $|t_n - t_m| < \varepsilon$ for all $n, m > N$. This implies that $|s_n - s_m| < \varepsilon$ thus s_n is convergent. ■

Theorem 6.6.3 (Ratio Test for Series)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Suppose $a_n \neq 0$ for all n and the sequence $\left(\frac{a_{n+1}}{a_n}\right)_{n\in\mathbb{N}}$ converges to some $l \in \mathbb{R}$. Then the following are true.

- If $0 \leq l < 1$, then $\sum_{k=0}^{\infty} a_k$ converges absolutely.
- If $l > 1$, then $\sum_{k=0}^{\infty} a_k$ diverges.

Proof
Since $\frac{a_{n+1}}{a_n} \to l < 1$, choose $\varepsilon > 0$ such that $l + \varepsilon < 1$. There exists an N such that $\left|\frac{a_{n+1}}{a_n} - l\right| < \varepsilon$ whenever $n > N$.

$$\begin{aligned}
\left|\frac{a_{n+1}}{a_n} - l\right| &< \varepsilon \\
\left|\left|\frac{a_{n+1}}{a_n}\right| - l\right| &< \varepsilon \\
\left|\frac{a_{n+1}}{a_n}\right| &< l + \varepsilon \\
|a_{n+1}| &< |a_n|(l + \varepsilon) \\
|a_{n+1}| &< |a_N|(l + \varepsilon)^{n+1-N}
\end{aligned}$$

Since

$$\sum_{k=1}^{\infty} |a_N|(l+\varepsilon)^{k+1-N} = \frac{a_N}{1-(l+\varepsilon)}$$

We have that $\sum_{k=1}^{\infty} |a_k|$ converges by comparison test.

Now suppose that $l > 1$, there exists N $|a_{n+1}| > |a_n|$ for some $n > N$. This means that (a_n) does not converge to 0 and by the null sequence test $\sum_{k=1}^{\infty} |a_k|$ diverges. ■

Theorem 6.6.4 (Ratio Test for Series)
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence. Suppose that the sequence $\left(|a_n|^{1/n}\right)$ converges to some $l \in \mathbb{R}$. Then the following are true.

- If $0 \leq l < 1$, then $\sum_{k=0}^{\infty} a_k$ converges absolutely.
- If $l > 1$, then $\sum_{k=0}^{\infty} a_k$ diverges.

6.7 Rearrangement of Series

Definition 6.7.1 (Rearrangement)
$(b_n)_{n\in\mathbb{N}}$ is a rearrangement of $(a_n)_{n\in\mathbb{N}}$ if there is a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that for all $n \in \mathbb{N}$ $b_n = a_{\sigma(n)}$

Proposition 6.7.2
Suppose $(b_n)_{n\in\mathbb{N}}$ is a rearrangement of $(a_n)_{n\in\mathbb{N}}$ and $a_n > 0$ for all n and $\sum_{k=1}^{\infty} a_k$ converges. Then $\sum_{k=1}^{\infty} b_k$ converges and

$$\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} a_k$$

Proof
Suppose that $s_n = \sum_{k=1}^{n} a_k$, $\sum_{k=1}^{\infty} a_k = s$ and $t_n = \sum_{k=1}^{n} b_k$. Suppose that $a_{\sigma(n)} = b_n$. Fix N and consider $M_N = \max\{\sigma(r) : r \leq N\}$. Then the first a_{M_n} terms contains the first b_n terms. Since $a_n, b_n > 0$ for all n, we have $t_n < s_{M_N} < s$. Thus t_n is increasing and bounded and converges by the completeness axiom to, say t with $t \leq s$. Reverse the roles of a_n and b_n now given that t_n converges, and we get $s \leq t$. Thus $s = t$. ■

Proposition 6.7.3
Suppose $(b_n)_{n\in\mathbb{N}}$ is a rearrangement of $(a_n)_{n\in\mathbb{N}}$ and $\sum_{k=1}^\infty a_k$ converges absolutely. Then $\sum_{k=1}^\infty b_k$ converges and $\sum_{k=1}^\infty b_k = \sum_{k=1}^\infty a_k$.

Proof
Let $\sum_{k=1}^\infty a_k$ converges absolutely and $\sum_{k=1}^\infty b_k$ a rearrangement of the series. Consider $u_n = \frac{1}{2}(|a_n| + a_n)$, $v_n = \frac{1}{2}(|a_n| - a_n)$, $x_n = \frac{1}{2}(|b_n| + b_n)$, $y_n = \frac{1}{2}(|b_n| - b_n)$. We have that

$$u_n = \begin{cases} a_n & a_n \geq 0 \\ 0 & a_n \leq 0 \end{cases}$$

$$v_n = \begin{cases} 0 & a_n \geq 0 \\ a_n & a_n \leq 0 \end{cases}$$

$$x_n = \begin{cases} b_n & b_n \geq 0 \\ 0 & b_n \leq 0 \end{cases}$$

$$y_n = \begin{cases} 0 & b_n \geq 0 \\ b_n & b_n \leq 0 \end{cases}$$

We have $\sum_{k=1}^\infty u_k \leq \sum_{k=1}^\infty |a_k|$ thus $\sum_{k=1}^\infty u_k$ is convergent by completeness axiom and

$$\sum_{k=1}^\infty -v_k \leq \sum_{k=1}^\infty |a_k|$$

thus $\sum_{k=1}^\infty -v_k$ converges by completeness axiom and

$$\sum_{k=1}^\infty -v_k = -\sum_{k=1}^\infty v_k$$

converges. We have that $\sum_{k=1}^\infty u_k = \sum_{k=1}^\infty x_k$ and $\sum_{k=1}^\infty v_k = \sum_{k=1}^\infty y_k$ by construction. Thus

$$\sum_{k=1}^\infty a_k = \sum_{k=1}^\infty (u_k - v_k) = \sum_{k=1}^\infty (x_k - y_k) = \sum_{k=1}^\infty b_k$$

is convergent. ■

Definition 6.7.4 (Conditionally Convergent)
A series $\sum_{k=1}^\infty a_k$ is said to be conditionally convergent if $\sum_{k=1}^\infty a_k$ is convergent but

$$\sum_{k=1}^\infty |a_k|$$

diverges.

Proposition 6.7.5
Suppose $\sum_{k=1}^\infty a_k$ is conditionally convergent. Consider $\sum_{k=1}^\infty u_k$ and $\sum_{k=1}^\infty v_k$ where

$$u_n = \begin{cases} a_n & \text{if } a_n \geq 0 \\ 0 & \text{if } a_n < 0 \end{cases}$$

and

$$v_n = \begin{cases} 0 & \text{if } a_n \geq 0 \\ a_n & \text{if } a_n < 0 \end{cases}$$

Then $\sum_{k=1}^\infty u_k = +\infty$ and $\sum_{k=1}^\infty v_k = -\infty$.

Proof
Note that $a_n = u_n - v_n$ and $|a_n| = u_n + v_n$. Suppose for contradiction that $\sum u_n$ converges. Then $|a_n| = 2u_n - a_n$ and thus $\sum |a_n|$ converges, a contradiction. Suppose also that $\sum v_n$ converges. Then again $|a_n| = 2v_n + a_n$ and thus $\sum |a_n|$ converges, a contradiction. Thus we have $\sum u_n$ and $\sum v_n$ diverges. Since $u_n \geq 0$ and $v_n \leq 0$ for all n, we have that $\sum u_n = \infty$ and $\sum v_n = -\infty$. ■

Theorem 6.7.6 (Riemann's Rearrangement Theorem)
Suppose $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. Then for every $l \in \mathbb{R}$ there exists a rearrangement (b_n) of (a_n) such that $\sum_{k=1}^{\infty} b_k = l$.

Proof
Let (p_n) be the subsequence of all positive terms of (a_n) and (q_n) all its negative terms. We have that $(p_n) \to +\infty$ and $(q_n) \to -\infty$. Consider $l \geq 0$. There exists N_1 such that $S_1 = \sum_{k=1}^{N_1} a_k < l$ and $S_1 + a_{N_1+1} > l$. Then since $(q_n) \to -\infty$. There exists M_1 such that $T_1 = S_1 + \sum_{k=1}^{M_1} q_k < l$ but $S_1 + \sum_{k=1}^{M_1-1} q_k > l$. Repeat the process to find the rearrangement of our sequence

$$p_1, \dots, p_{N_1}, q_1, \dots, q_{M_1}, p_{N_1+1}, \dots, p_{N_2}, q_{M_1+1}, \dots, q_{M_2}, \dots$$

Its partial sums converges since $|S_i - l| \leq p_{N_i}$ and $|T_i - l| \leq -q_{M_i}$ for all i and p_{N_i} and q_{M_i} tends to 0 since a_n is null. Swap the roles of q_n and p_n for the case $l < 0$. ■

7 Sequences and Series of Functions

7.1 Uniform Convergence

Definition 7.1.1 (Pointwise Convergence)
Suppose $(f_n)_{n\in\mathbb{N}}$ is a sequence of functions defined on an interval I. Suppose that the sequence of numbers $(f_n(x))_{n\in\mathbb{N}}$ converges for every $x \in I$. We can define

$$f(x) = \lim_{n\to\infty} f_n(x)$$

for all $x \in I$. We say that $(f_n)_{n\in\mathbb{N}}$ converges to f.

Definition 7.1.2 (Uniform Convergence)
We say that $(f_n)_{n\in\mathbb{N}}$ converges uniformly on I to a function f if for every $\varepsilon > 0$ there is an integer N such that $n > N$ implies

$$|f_n(x) - f(x)| < \varepsilon$$

for all $x \in I$.

Example 7.1.3
Define

$$f_n(x) = \frac{x}{n}$$

for $n \in \mathbb{N}$. The sequence $(f_n)_{n\in\mathbb{N}}$ is not uniformly convergent.

Example 7.1.4
Define

$$f_n(x) = \frac{x}{1+x^n}$$

for $n \in \mathbb{N}$. The sequence $(f_n)_{n\in\mathbb{N}}$ is not uniformly convergent.

Corollary 7.1.5
Uniform convergence implies pointwise convergence.

Proof
Suppose that $(f_n)_{n\in\mathbb{N}}$ is uniformly convergent to f. Then in particular fix any x in its domain, its epsilon-delta definition of convergence on x is precisely the definition of pointwise convergence. ∎

Theorem 7.1.6 (Cauchy Criterion)
The sequence of functions $(f_n)_{n\in\mathbb{N}}$ defined on I converges uniformly if and only if for every $\varepsilon > 0$ there eixsts N such that $m, n > N$ implies

$$|f_m(x) - f_n(x)| < \varepsilon$$

for all x in the domain.

Proof
Suppose that $(f_n)_{n\in\mathbb{N}}$ converges uniformly to f. Then fix $\frac{\varepsilon}{2} > 0$, there exists $N \in \mathbb{N}$ such that $n > N$ implies $|f_n - f| < \frac{\varepsilon}{2}$ and $|f_m - f| < \frac{\varepsilon}{2}$ if $m > N$. Then

$$|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f_n(x) - f(x)| < \varepsilon$$

Now suppose that the cauchy criterion is satisfied. Then for every fixed x, $f_n(x)$ is Cauchy in $\mathbb{R}$ and thus is convergent. This means there exists f on its domain such that at least f_n is pointwise convergent to f. From the cauchy criterion, we have that

$$f_m(x) - \varepsilon < f_n(x) < f_m(x) + \varepsilon$$

for all x and $n, m > N$. Since the bounds hold for all $m > N$, and we have that $f_m(x) \to f$ pointwise, we have that

$$f(x) - \varepsilon < f_n(x) < f(x) + \varepsilon$$

and thus

$$|f(x) - f_n(x)| < 2\varepsilon$$

for all $n > N$. ■

Side note: Since we want a universal N such that it works for every $x \in I$, we can simply consider the maximum of the difference between $f_m(x)$ and $f_n(x)$, which leads to some notes developing the norm of a functionn $\|f\|_\infty = \sup_{x\in I} |f(x)|$. They are essentially the same even when substitued in the definition.

Proposition 7.1.7
Suppose $\lim_{n\to\infty} f_n(x) = f(x)$. Let

$$\sup_{x\in I} |f_n(x) - f(x)| = \|f_n - f\|_\infty$$

Then $(f_n)_{n\in\mathbb{N}}$ converges to f uniformly if and only if

$$\lim_{n\to\infty} \|f_n - f\|_\infty = 0$$

Proof
Suppose that $(f_n)_{n\in\mathbb{N}}$ converges to f uniformly. Then trivially $\sup_{x\in I} |f_n - f| < \varepsilon$. Thus we are done.

Suppose that $\|f_n - f\| \to 0$ as $n \to \infty$. Then since $|f_n - f| \leq \|f_n - f\|_\infty < \varepsilon$ in the ε definition, we must have $(f_n)_{n\in\mathbb{N}}$ converging to f uniformly. ■

7.2 Uniform Convergence and Continuity

Theorem 7.2.1
Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of continuous functions on I, and if f_n converges to f uniformly on I, then f is continuous on I.

Proof
By uniform convergence, we know that if $\frac{\varepsilon}{3} > 0$, there exists $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$ for all $x \in I$ and $n > N$. Fix n such that $n > N$ now. I will show that f is continuous at x_0. Since $f_n(x)$ is continuous at x_0, we know that there exists δ such that $x \in (x_0 - \delta, x_0 + \delta) \cap I$ implies $|f_n(x) - f_n(x_0)| < \frac{\varepsilon}{3}$. Now

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon \end{aligned}$$

Thus f is continuous at x_0. ■

Proposition 7.2.2
Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of continuous and bounded functions on $I \subseteq \mathbb{R}$ that is cauchy. Then it converges to a continuous and bounded function.

Proof
We alredy showed that uniform convergence implies continuity on its limit, we just have to show that it is bounded. Note that for all $x \in I$, the domain of f,

$$|f(x)| \leq |f(x) - f_n(x)| + |f_n(x)|$$

By uniformly convergence, there exists $N \in \mathbb{N}$ such that $n > N$ implies $|f_n(x) - f(x)| < 1$. For that n, since f_n is bounded, we have $|f_n| < M$ is f_n is bounded. Thus we have that $|f(x)| \leq M + 1$ for all $x \in I$. ■

7.3 Uniform Convergence and Integrability

Theorem 7.3.1
Suppose $(f_n)_{n\in\mathbb{N}}$ is a sequence of functions that is Riemann Integrable on $[a, b]$ that converges uniformly on $[a, b]$ to a function f. Then f is Riemann Integrable and

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b \lim_{n\to\infty} f_n(x)\,dx$$

Proof
We first show that f is integrable. Fix $\frac{\varepsilon}{4(b-a)} > 0$. Then there exists $N \in \mathbb{N}$ such that $n > N$ implies

$$|f_n(x) - f(x)| < \frac{\varepsilon}{4(b-a)}$$

for all x. Fix $n > N$. Since f_n is integrable, given $\frac{\varepsilon}{2} > 0$ there exists a partition such that $U(f_n, P) - L(f_n, P) < \frac{\varepsilon}{2}$. For that P,

$$\begin{aligned}
U(f, P) - L(f, P) &= \sum_{k=1}^{n} (\sup_{I_k} f - \inf_{I_k} f)|I_k| \\
&= \sum_{k=1}^{n} (\sup_{I_k}(f - f_n + f_n) - \inf_{I_k}(f - f_n + f_n))|I_k| \\
&\leq \sum_{k=1}^{n} \left(\sup_{I_k} |f - f_n| + \sup_{I_k} f_n + \inf_{I_k} |f - f_n| - \inf_{I_k} f_n \right) |I_k| \\
&= 2\sum_{k=1}^{n} \|f_n - f\|_\infty |I_k| + \sum_{k=1}^{n} (\sup_{I_k} f_n - \inf_{I_k} f_n)|I_k| \\
&\leq 2\|f_n - f\|_\infty (b-a) + U(f_n, P) - L(f_n, P) \\
&\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon
\end{aligned}$$

Thus we have that f is integrable.

Now note that

$$\begin{aligned}
\left| \int_a^b f_n - \int_a^b f \right| &= \left| \int_a^b (f_n - f) \right| \\
&\leq \int_a^b |f_n - f| \\
&\leq \int_a^b \|f - f_n\|_\infty \\
&= \|f_n - f\|_\infty (b-a)
\end{aligned}$$

which goes to 0 as $n \to \infty$ by the uniform convergence of f_n to f. ■

Example 7.3.2

Define

$$f_n(x) = \begin{cases} n & \text{if } 0 \leq x \leq \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

for $n \in \mathbb{N} \setminus \{0\}$. The the following are true.
- The sequence $(f_n)_{n\in\mathbb{N}\setminus\{0\}}$ is not uniformly convergent.
- $\lim_{n\to\infty} \int_0^1 f_n(x) \neq \int_0^1 \lim_{n\to\infty} f_n(x)$.

This example show that without uniform convergence, the limit and the integration opeartor cannot be interchanged.

7.4 Uniform Convergence and Differentiability

Theorem 7.4.1
Suppose $(f_n)_{n\in\mathbb{N}} \subset C^1([a,b])$ and it is pointwise convergent to f. If $(f_n')_{n\in\mathbb{N}}$ converges uniformly on $[a,b]$, then $f \in C^1([a,b])$ and $(f_n')_{n\in\mathbb{N}}$ in fact converges uniformly to f'. In other words, we have

$$\lim_{n\to\infty}\left(\frac{d}{dx}f_n(x)\right) = \frac{d}{dx}\left(\lim_{n\to\infty} f_n(x)\right)$$

Proof
Suppose that f_n' converges uniformly to g. By 7.3.1, we have that

$$\int_a^x g(t)\,dt = \int_a^x \lim_{n\to\infty} f_n'(t)\,dt = \lim_{n\to\infty}\int_a^x f_n'(t)\,dt$$

By the FTC, this gives

$$\int_a^x g(t)\,dt = \lim_{n\to\infty}(f_n(x) - f_n(a)) = f(x) - f(a)$$

Now since $\{f_n\}$ is continuous thus g is continuous. By the above, we also have that f is continuous. Thus the FTC implies that $\frac{d}{dx}\int_a^x g(t)\,dt = g(x)$. Thus we have that $g = f'$. ■

An easy way to remember this is $(f_n)_{n\in\mathbb{N}}$ pointwise convergent, $(f_n')_{n\in\mathbb{N}}$ are all continuous and converges uniformly, we can exchange the order of differentiation and limit.

Example 7.4.2
Define

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}$$

for $n \in \mathbb{N} \setminus \{0\}$. Then the following are true.
- f_n is continuously differentiable.
- $(f_n)_{n\in\mathbb{N}\setminus\{0\}}$ converges uniformly to $|x|$.
- $\lim_{n\to\infty} \frac{d}{dx} f_n(x) \neq \frac{d}{dx} \lim_{n\to\infty} f_n(x)$.

This example show that without uniform convergence of the derivatives of $f_n(x)$, the limit and the differentiation opeartor cannot be interchanged.

Example 7.4.3
Define

$$f_n(x) = \frac{1}{n}\sin\left(n^2 x\right)$$

for $n \in \mathbb{N}$. Then the following are true.
- f_n is continuously differentiable.
- $(f_n)_{n\in\mathbb{N}\setminus\{0\}}$ converges uniformly.
- $\lim_{n\to\infty} \frac{d}{dx} f_n(x) \neq \frac{d}{dx} \lim_{n\to\infty} f_n(x)$.

This example show that without uniform convergence of the derivatives of $f_n(x)$, the limit and the differentiation opeartor cannot be interchanged.

7.5 Series of Functions

Definition 7.5.1 (Pointwise Convergence of Series of Functions)
Let $(f_k)_{k\in\mathbb{N}}$ be a sequence of functions $f_k : [a, b] \to \mathbb{R}$. Define

$$S_n(x) = \sum_{k=1}^{n} f_k(x)$$

Then the series converges pointwise to $S : [a, b] \to \mathbb{R}$ if $S_n \to S$ pointwise.

Definition 7.5.2 (Uniform Convergence of Series of Functions)
Let $(f_k)_{k\in\mathbb{N}}$ be a sequence of functions $f_k : [a, b] \to \mathbb{R}$. Define

$$S_n(x) = \sum_{k=1}^{n} f_k(x)$$

Then the series converges uniformly to $S : [a, b] \to \mathbb{R}$ if S_n converges to S uniformly.

Theorem 7.5.3
Let $(f_n)_{n\in\mathbb{N}}$ such that $f_n : [a, b] \to \mathbb{R}$ is a sequence of Riemann Integrable functions. If $S_n = \sum_{k=1}^{n} f_k$ converges uniformly, then $\sum_{k=1}^{\infty} f_k$ is Riemann Integrable and

$$\int_a^b \sum_{k=1}^{\infty} f_k(x)\,dx = \sum_{k=1}^{\infty} \int_a^b f_k(x)\,dx$$

Proof
Since S_n is a finite sum of integrable functions, S_n is also integrable by additivity. By theorem 7.3.1, S is also integrable and we have

$$\lim_{n\to\infty} \int S_n = \int \lim_{n\to\infty} S_n$$

Substituting $S_n = \sum_{k=1}^{n} f_k$ and we get the result. ■

Theorem 7.5.4 (Term by Term Differentiation)
Let $(f_n)_{n\in\mathbb{N}}$ such that $f_n : [a, b] \to \mathbb{R}$ is a sequence of C^1 functions. If $S_n = \sum_{k=1}^{n} f_k$ converges pointwise, and $\sum_{k=1}^{n} f_k'$ converges uniformly, then

$$\left(\sum_{k=1}^{\infty} f_k(x)\right)' = \sum_{k=1}^{\infty} f_k'(x)$$

Proof
Similar to the proof above, except that it utilizes theorem 7.4.1 instead. ■

Theorem 7.5.5 (Weierstrass M-test)
Let $\{f_n\}$ be a sequence of functions $f_n : [a, b] \to \mathbb{R}$, and assume that for every n there exists $M_n > 0$ such that $|f_n(x)| \leq M_n$ for every $x \in [a, b]$ and $\sum_{k=1}^{\infty} M_k$ is finite. Then

$$S_n = \sum_{k=1}^{n} f_k$$

converges uniformly on $[a, b]$ to the function $\sum_{k=1}^{\infty} f_k$.

Proof
Since $\sum_{k=1}^{n} M_k < \infty$, given $\varepsilon > 0$ there exists N such that

$$\sum_{k=m+1}^{n} M_k < \varepsilon$$

for all $m, n > N$ by cauchy condition of infinite sums. Now note that

$$\begin{aligned} |S_n(x) - S_m(x)| &= \left| \sum_{k=m+1}^{n} f_k(x) \right| \\ &\leq \sum_{k=m+1}^{n} |f_k(x)| \\ &\leq \sum_{k=m+1}^{n} M_k \\ &< \varepsilon \end{aligned}$$

This proves that S_n is uniformly cauchy, which implies that S_n is uniformly convergent. ■

8 Power Series and Taylor Series

8.1 Power Series

Definition 8.1.1 (Power Series)
A power series in $\mathbb{R}$ is an infinite series of the form

$$\sum_{k=0}^{\infty} a_k(x-c)^k$$

where $c \in \mathbb{R}$ is the center and $a_k \in \mathbb{R}$ for all $k \in \mathbb{N}$.

We treat the case of when $c = 0$ so that the power series is centered at 0. The more general case follows immediately by a translation.

Proposition 8.1.2
Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with $\sum_{k=0}^{\infty} a_k t^k$ convergent for some $t \in \mathbb{R}$. Then $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely for all x with $|x| < |t|$.

Proof
Note that the convergence of $\sum_{k=0}^{\infty} a_k t^k$ implies $(a_n t^n) \to 0$. This implies that $a_n t^n$ is bounded by, say M.

$$\begin{aligned}\sum_{k=0}^{\infty} |a_k x^k| &= \sum_{k=0}^{\infty} |a_k t^k| \left|\frac{x^k}{t^k}\right| \\ &< \sum_{k=0}^{\infty} M \left|\frac{x^k}{t^k}\right| \\ &= \frac{M}{1 - \left|\frac{x}{t}\right|}\end{aligned}$$

Since it is increasing and bounded, the infinity sum converges absolutely thus the sum converges. ■

Definition 8.1.3 (Radius of Convergence)
Let $\sum_{k=0}^{\infty} a_k(x-c)^k$ be a power series. Define the radius of convergence of the power series to be

$$R = \sup\left\{ r \in \mathbb{R} \;\middle|\; \sum_{k=0}^{\infty} a_k r^k \text{ converges} \right\}$$

Proposition 8.1.4
Let $\sum_{k=0}^{\infty} a_k x^k$ with radius of convergence R. Then $\sum_{k=0}^{\infty} |a_k| x^k$ also has radius of convergence R.

Proof
We have that

$$\left|\sum_{k=0}^{\infty} |a_k| x^k\right| \leq \sum_{k=0}^{\infty} |a_k||x^k|$$

which converges between $(-R, R)$. ■

Theorem 8.1.5

For any power series $\sum_{k=0}^{\infty} a_k x^k$, the radius of convergence is

$$R = \frac{1}{\limsup_{n\to\infty} |a_n|^{\frac{1}{n}}}$$

Proof

Application of the root test.

Proposition 8.1.6 (Continuity of Power Series)

Let $\sum_{k=0}^{\infty} a_k x^k$ with radius of convergence R. Then the function $f(x) = \sum_{k=0}^{\infty} a_k x^k$ is continuous on the interval $(-R, R)$.

Proof

Suppose $|x| < R$. We show that the function is continuous at x. Let T such that $|x| < T < R$. Then $\sum_{k=0}^{\infty} |a_k| T^k$ converges, so for every $\varepsilon > 0$ there is some number N such that

$$\sum_{k=N+1}^{\infty} |a_k| T^k < \frac{\varepsilon}{3}$$

Let y such that $|y - x| < T - |x|$. Then we will have $|y| < T$ and $|x| < T$. Hence

$$\sum_{k=N+1}^{\infty} |a_k||x|^k < \frac{\varepsilon}{3}$$

and

$$\sum_{k=N+1}^{\infty} |a_k||y|^k < \frac{\varepsilon}{3}$$

The partial sum

$$\sum_{k=0}^{N} a_k y^k$$

is a polynomial thus is continuous. There exists some $\delta_0 > 0$ such that $|y - x| < \delta_0$ implies

$$\left| \sum_{k=0}^{N} a_k y^k - \sum_{k=0}^{N} a_k x^k \right| < \frac{\varepsilon}{3}$$

Choose $\delta = \min(\delta_0, T - |x|)$. Then we have that $|y - x| < \delta$ we get

$$\begin{aligned}
\left| \sum_{k=0}^{\infty} a_k y^k - \sum_{k=0}^{\infty} a_k x^k \right| &\leq \left| \sum_{k=N+1}^{\infty} a_k y^k \right| + \left| \sum_{k=0}^{N} a_k y^k - \sum_{k=0}^{N} a_k x^k \right| + \left| \sum_{k=N+1}^{\infty} a_k x^k \right| \\
&\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\
&= \varepsilon
\end{aligned}$$

Proposition 8.1.7 (Differentiability of Power Series)

Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with radius of convergence R. Then f is differentiable on $(-R, R)$ and

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

Proof

We first show that $\sum_{k=1}^{\infty} k a_k x^{k-1}$ has the same radius of convergence. We have that $\sum_{k=0}^{\infty} |a_k| x^k$

has the same radius of convergence. Choose x and y such that $x < y < R$. Then $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} a_k y^k$ converges.

$$\begin{aligned}\sum_{k=1}^{\infty} k a_k x^{k-1} &< \sum_{k=1}^{\infty} |a_k|(y^{k-1} + y^{k-2}x + \cdots + x^{k-1}) \\ &= \sum_{k=0}^{\infty} |a_k| \frac{y^k - x^k}{y - x}\end{aligned}$$

Since $\sum_{k=0}^{\infty} |a_k| \frac{y^k - x^k}{y-x}$ converges, we complete our lemma.

Now for the main proof. Choose T such that $|x| < T < R$. We have proved that $\sum_{k=1}^{\infty} k a_k x^{k-1}$ converges so given $\varepsilon > 0$ there is a number N so that

$$\sum_{k=N+1}^{\infty} k|a_k|T^{k-1} < \frac{\varepsilon}{3}$$

Now if $0 < |y - x| < T - |x|$, we have $|y| < T$ and $|x| < T$. Thus we have that

$$\left| \sum_{k=N+1}^{\infty} k a_k x^{k-1} \leq \sum_{k=N+1}^{\infty} k|a_k||x|^{k-1} < \frac{\varepsilon}{3} \right|$$

and also

$$\begin{aligned}\left| \sum_{k=N+1}^{\infty} a_k \frac{y^k - x^k}{y - x} \right| &= \left| \sum_{k=N+1}^{\infty} a_k (y^{k-1} + y^{k-2}x + \cdots + x^{k-1}) \right| \\ &\leq \sum_{k=N+1}^{\infty} |a_k|(|y|^{k-1} + \cdots + |x|^{k-1}) \\ &\leq \sum_{k=N+1}^{\infty} k|a_k|T^{k-1} \\ &\leq \frac{\varepsilon}{3}\end{aligned}$$

The sum

$$\sum_{k=1}^{N} a_k (y^{k-1} + y^{k-2}x + \cdots + x^{k-1})$$

is a polynomial in y whose value at x is $\sum_{k=1}^{N} k a_k x^{k-1}$ so there exists $\delta_0 > 0$ such that $|y - x| < \delta_0$ implies

$$\begin{aligned}\left| \sum_{k=1}^{N} a_k \frac{y^k - x^k}{y - x} - \sum_{k=1}^{N} \right| k a_k x^{k-1} &= \left| \sum_{k=1}^{N} a_k (y^{k-1} + y^{k-2}x + \cdots + x^{k-1}) - \sum_{k=1}^{N} k a_k x^{k-1} \right| \\ &\leq \frac{\varepsilon}{3}\end{aligned}$$

Finally, choose $\delta = \min(\delta_0, T - |x|)$ then $|y - x| < \delta$ implies

$$\begin{aligned}\left| \sum_{k=1}^{\infty} a_k \frac{y^k - x^k}{y - x} - \sum_{k=1}^{\infty} k a_k x^{k-1} \right| &\leq \left| \sum_{k=N+1}^{\infty} a_k \frac{y^k - x^k}{y - x} \right| + \left| \sum_{k=1}^{N} a_k \frac{y^k - x^k}{y - x} - \sum_{k=1}^{N} k a_k x^{k-1} \right| \\ &+ \left| \sum_{k=N+1}^{\infty} k a_k x^{k-1} \right| \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon\end{aligned}$$

■

Lemma 8.1.8
Every power series is a smooth function.

8.2 Analytic Functions and Taylor Series

Given a function $f : U \to \mathbb{R}$, we may ask whether f can be represented as a power series. Since power series are infinitely differentiable, we expect any function representable by a power series to also be infinitely differentiable. However there are more conditions needed for f to representable by a power series.

Definition 8.2.1 (Analytic Functions)
Let $f : U \subseteq \mathbb{R} \to \mathbb{R}$ be a function. Let $(a, b) \subseteq U$ be an open interval. We say that f is analytic on (a, b) if there exists $a_k \in \mathbb{R}$ for each $k \in \mathbb{N}$ and $c \in (a, b)$ such that f can be expressed as a power series

$$f(x) = \sum_{k=0}^{\infty} a_k (x-c)^k$$

for all $x \in (a, b)$.

Proposition 8.2.2
Let $f : U \subseteq \mathbb{R} \to \mathbb{R}$ be a function. Then the following are true.
- If f is analytic at $(a, b) \subseteq U$, then $f \in \mathcal{C}^\infty(a, b)$.
- Moreover, if the power series representation of f at (a, b) is given by

  $$f(x) = \sum_{k=0}^{\infty} a_k (x-c)^k$$

  for some $c \in (a, b)$ and $a_k \in \mathbb{R}$, then we have

  $$a_k = \frac{f^{(k)}(c)}{k!}$$

Smooth functions may not be analytic in general. Consider the function

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

It is smooth but is not analytic at any interval containing $x = 0$.

Definition 8.2.3 (Taylor Series)
Let $f : (a, b) \to \mathbb{R}$ be a smooth function. Let $c \in (a, b)$. Define the Taylor series of f to be

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

Definition 8.2.4 (Remainder)
Let $f : (a, b) \to \mathbb{R}$ in C^n. Let $c \in (a, b)$. Then define the nth remainder of f at c to be

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(x-c)^{n-1}$$

Theorem 8.2.5
Let $f : (a, b) \to \mathbb{R}$ possess derivatives of all order at $c \in (a, b)$. Then

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

if and only if

$$\lim_{n\to\infty} R_n(x) = 0$$

Proof
Suppose that $f(x)$ is equal to its taylor series at c. Since the taylor series converges to $f(x)$, $R_n(x)$ converges to 0 by definition.

Suppose that $R_n(x)$ converges to 0. Then by the definition of $R_n(x)$ we must have $f(x)$ equal to its taylor and we are done. ■

Theorem 8.2.6 (Lagrange Remainder)
If $f : (c, d) \to \mathbb{R}$ is n times differentiable and $a, b \in (c, d)$, then

$$R_n(b) = \frac{f^{(n)}(t)}{n!}(b-a)^n$$

for some t between a and b.

Proof
Define

$$g(x) = f(x) - \left(f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(x-a)^{n-1}\right)$$

$g(x)$ satisfies $g(a) = g'(a) = \cdots = g^{(n-1)}(a) = 0$. It also satisfies $g^{(n)}(x) = f^{(n)}(x)$ for all x. Define $h(x) = g(x) - g(b)\frac{(x-a)^n}{(b-a)^n}$. Then $h(a) = h'(a) = \cdots = h^{(n-1)}(a) = 0$ and $h(b) = 0$. Now since $h(a) = h(b) = 0$, there exists $t_1 \in (a, b)$ where $h'(t_1) = 0$ by Rolle's Theorem. Since $h'(t_1) = h'(a) = 0$, there exists $t_2 \in (a, t_1)$ where $h''(t_2) = 0$. Continuing this way we get that $t = t_n$ where $h^{(n)}(t) = 0$. In terms of g, we have

$$g^{(n)}(t) = g(b)\frac{n!}{(b-a)^n}$$

Rewriting the equation we have that

$$g(b) = \frac{g^{(n)}(t)}{n!}(b-a)^n = \frac{f^{(n)}(t)}{n!}(b-a)^n$$

Substituting $x = b$ into the definition of $g(x)$, we have

$$\begin{aligned}
g(b) &= f(b) - \left(f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1}\right) \\
\frac{f^{(n)}(t)}{n!}(b-a)^n &= f(b) - \left(f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1}\right) \\
&= f(b) - \sum_{k=0}^{n-1} \frac{f^{(k)(a)}}{(k-1)!}(b-a)^{k-1} \\
&= R_n(b)
\end{aligned}$$

Thus we are done. ■

8.3 The Trigonometric Series

Lemma 8.3.1
The radius of convergence of the two power series

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} \quad \text{and} \quad \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

is ∞.

Proof
Let $a_k = \frac{(-1)^k x^{2k+1}}{(2k+1)!}$. Then we have $\left|\frac{a_{k+1}}{a_k}\right| = \frac{x^{2k+1}}{(2k+3)(2k+2)} \leq \frac{x^2}{k^2}$. Hence the ratio of the two terms converge to 0. By the ratio test for series, the power series converges absolutely and hence converges for all $x \in \mathbb{R}$.

Similarly, let $b_n = \frac{(-1)^k x^{2k}}{(2k)!}$. Then we have $\left|\frac{b_{k+1}}{b_k}\right| = \frac{x^2}{(2k+2)(2k+1)} \leq \frac{x^2}{k^2}$. Hence the ratio of the two terms converge to 0. By the ratio test for series, the power series converges absolutely and hence converges for all $x \in \mathbb{R}$.

■

Definition 8.3.2 (Sine and Cosine)
Define the sine function $\sin : \mathbb{R} \to \mathbb{R}$ to be the power series

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

Define the cosine function $\cos : \mathbb{R} \to \mathbb{R}$ to be the power series

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

Lemma 8.3.3
The following are true regarding sine and cosine.
- $\frac{d}{dx}\sin(x) = \cos(x)$.
- $\frac{d}{dx}\cos(x) = -\sin(x)$.

Proof
We have

$$\begin{aligned}
\frac{d}{dx}\sin(x) &= \frac{d}{dx}\left(\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}\right) \\
&= \sum_{k=0}^{\infty} \frac{d}{dx}\left(\frac{(-1)^k x^{2k+1}}{(2k+1)!}\right) \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) x^{2k}}{(2k+1)!} \\
&= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \\
&= \cos(x)
\end{aligned}$$

and

$$\begin{aligned}
\frac{d}{dx}\cos(x) &= \frac{d}{dx}\left(\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k}}{(2k)!}\right) \\
&= \sum_{k=0}^{\infty}\frac{d}{dx}\left(\frac{(-1)^k x^{2k}}{(2k)!}\right) \\
&= \sum_{k=1}^{\infty}\frac{(-1)^k(2k)x^{2k-1}}{(2k)!} \\
&= \sum_{k=1}^{\infty}\frac{(-1)^k x^{2k-1}}{(2k-1)!} \\
&= -\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k+1}}{(2k+1)!} \\
&= -\sin(x)
\end{aligned}$$

■

Lemma 8.3.4
The following are true regarding sine and cosine.
- Pythagorean Identity: $\sin^2(x)+\cos^2(x)=1$ for all $x\in\mathbb{R}$.
- Addition law for sine: $\sin(x+y)=\sin(x)\cos(y)+\cos(x)\sin(y)$ for all $x,y\in\mathbb{R}$.
- Addition law for cosine: $\cos(x+y)=\cos(x)\cos(y)-\sin(x)\sin(y)$ for all $x,y\in\mathbb{R}$.

Proof
Let $f(x)=\sin^2(x)+\cos^2(x)$. Then $f'(x)=2\sin(x)\cos(x)-2\cos(x)\sin(x)=0$. Hence $f(x)=k$ for some constant $k\in\mathbb{R}$. Then $k=f(0)=\sin^2(0)+\cos^2(0)=1$.

Let $y\in\mathbb{R}$ be fixed. Define

$$g(x)=\sin(x+y)-(\sin(x)\cos(y)+\cos(x)\sin(y))$$

and

$$h(x)=\cos(x+y)-(\cos(x)\cos(y)-\sin(x)\sin(y))$$

Then we compute that

$$g'(x)=\cos(x+y)-(\cos(x)\cos(y)-\sin(x)\sin(y))=h(x)$$

and

$$h'(x)=-\sin(x+y)-(-\sin(x)\cos(y)-\cos(x)\sin(y))=-g(x)$$

Let $S(x)=g^2(x)+h^2(x)$. Then we have

$$\begin{aligned}
S'(x) &= 2g(x)g'(x)+2h(x)h'(x) \\
&= 2g(x)h(x)-2h(x)g(x) \\
&= 0
\end{aligned}$$

Hence $S(x)=k$ for some constant $k\in\mathbb{R}$. Now since $g(0)=h(0)=0$, we have $S(0)=0$ hence $g^2(x)+h^2(x)=S(x)=0$ for all $x\in\mathbb{R}$. Since g^2 and h^2 are non-negative, we must have $g=0$ and $h=0$, hence the addition laws follows. ■

Proposition 8.3.5
There exists $x>0$ such that $\cos(x)=0$.

Proof
By definition of cosine, we have $\cos(1) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}$. I claim that $\frac{1}{2} < \cos(1) < \frac{13}{24}$. Let $S_n = \sum_{k=0}^{n} \frac{(-1)^k}{(2k)!}$. I prove that $S_n > 1/2$ for all $n \geq 1$. We proceed by induction. The base case is clear. Now suppose that $S_i > 1/2$ for all $1 \leq i \leq n$. If n is odd, then $S_{n+1} - S_n = \frac{(-1)^{n+1}}{(2n+2)!} > 0$. Hence $S_{n+1} > S_n > 1/2$. If n is even, then notice that $S_{n+1} = S_{n-1} + \frac{(-1)^n}{(2n)!} + \frac{(-1)^{n+1}}{(2n+2)!}$ and $\frac{(-1)^n}{(2n)!} > \frac{(-1)^{n+1}}{(2n+2)!}$ implies that $S_{n+1} > S_{n-1} > 1/2$. Thus the induction is complete. Since $(S_n)_{n\in\mathbb{N}} \to \cos(1)$ and $S_n > 1/2$ for $n \geq 1$, we must have $\cos(1) > 1/2$.

By a similar argument, we use induction to show that $S_n < S_3 = \frac{13}{24}$ so that $\cos(1) < \frac{13}{24}$.

By the addition law for cosine, we have

$$\cos(2) = \cos(1+1) = \cos^2(1) - \sin^2(1) = \cos^2(1) - 1 + \cos^2(1) = 2\cos^2(1) - 1 < 2 \cdot \frac{169}{576} - 1 < 0$$

By the IVT, we conclude that $\cos(x) = 0$ has a solution between 1 and 2. ■

Definition 8.3.6 (The Constant π)
Define the constant π to be the smallest positive real number such that $\cos\left(\frac{\pi}{2}\right) = 0$.

Lemma 8.3.7
The following are true regarding the values of sine and cosine.

	$x=0$	$x=\frac{\pi}{2}$	$x=\pi$	$x=\frac{3\pi}{2}$	$x=2\pi$
$\sin(x)$	0	1	0	-1	0
$\cos(x)$	1	0	-1	0	1

Proof
Clearly $\sin(0) = 0$ and $\cos(0) = 1$ by definition of sine and cosine.

By the Pythagorean identity, we have $\sin^2(\pi/2) = 1$, and so $\sin(\pi/2) = \pm 1$. Now since $\pi/2$ is the smallest positive root of $\cos(x)$, we must have that $\cos(x) > 0$ for $0 < x < \pi/2$. Hence $\frac{d}{dx}\sin(x) > 0$ for $0 < x < \pi/2$. This proves that $\sin(x)$ is strictly increasing for $0 < x < \pi/2$. Since $\sin(0) = 0$, we must have $\sin(\pi/2) = 1$.

By the addition law for sine, we have $\sin(\pi) = \sin\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = 2\sin(\pi/2)\cos(\pi/2) = 0$. Similarly, we have $\cos(\pi) = \cos^2(\pi/2) - \sin^2(\pi/2) = -1$.

By the addition law for sine, we have $\sin(3\pi/2) = \sin(\pi + \pi/2) = \sin(\pi)\cos(\pi/2) + \cos(\pi)\sin(\pi/2) = -1$. Similarly, we have $\cos(3\pi/2) = \cos(\pi + \pi/2) = \cos(\pi)\cos(\pi/2) - \sin(\pi)\sin(\pi/2) = 0$.

Finally, we have $\sin(2\pi) = 2\sin(\pi)\cos(\pi) = 0$ and $\cos(2\pi) = \cos^2(\pi) - \sin^2(\pi) = 1$. ■

Proposition 8.3.8
The following are true regarding sine and cosine.

- $\sin(x + 2\pi) = \sin(x)$ for all $x \in \mathbb{R}$.
- $\cos(x + 2\pi) = \cos(x)$ for all $x \in \mathbb{R}$.
- $\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$ for all $x \in \mathbb{R}$.

Proof
By the addition law for sine, we have $\sin(x + 2\pi) = \sin(x)\cos(2\pi) + \cos(x)\sin(2\pi) = \sin(x)$. Similarly, we have $\cos(x + 2\pi) = \cos(x)\cos(2\pi) - \sin(x)\sin(2\pi) = \cos(x)$.

By the addition law for sine, we have $\sin(x + \pi/2) = \sin(x)\cos(\pi/2) + \cos(x)\sin(\pi/2) = \cos(x)$. ∎

Corollary 8.3.9
The following are true regarding sine and cosine.
- $\sin(x)$ is strictly increasing in $[0, \pi/2) \cup [3\pi/2, 2\pi)$ and strictly decreasing in $[\pi/2, 3\pi/2)$.
- $\cos(x)$ is strictly increasing in $[\pi, 2\pi)$ and strictly decreasing in $[0, \pi)$.

Proof
We have seen that $\cos(x) > 0$ for $x \in [0, \pi/2)$. Then $\frac{d}{dx}\sin(x) = \cos(x)$ implies that $\sin(x)$ is strictly increasing in $[0, \pi/2)$. Since $\sin(x + \pi/2) = \cos(x)$ and $\sin(x)$ is 2π-periodic, we have that $\cos(x)$ is strictly increasing in $[3\pi/2, 2\pi)$. Since $\sin(0) = 0$, we must have $\sin(x) > 0$ for $[0, \pi/2)$ so that $-\cos(x)$ is strictly increasing in that interval, and so $\cos(x)$ is strictly decreasing in that interval. Again since $\sin(x + \pi/2) = \cos(x)$, we conclude that $\sin(x)$ is strictly increasing in $[3\pi/2, 2\pi)$.

Since $\sin(x + \pi/2) = \cos(x)$ and $\cos(x)$ is strictly decreasing in $[0, \pi/2)$, we must have that $\sin(x)$ is strictly decreasing in $[\pi/2, \pi)$. Since $\sin(\pi) = 0$, we must have that $\sin(x) > 0$ in that interval. Therefore $-\cos(x)$ is strictly decreasing in that interval, and that $\cos(x)$ is strictly increasing in that interval. Then $\sin(x + \pi/2) = \cos(x)$ implies that $\sin(x)$ is strictly increasing in $[\pi, 3\pi/2)$. Since $\sin(3\pi/2) = 0$, we must have $\sin(x) < 0$ in $[\pi, 3\pi/2)$. Therefore $-\cos(x)$ is decreasing, and finally that $\cos(x)$ is increasing in that interval. ∎

More work remains to prove that π is equal to the ratio of circumference to diameter, as known classically.

8.4 The Exponential Series

Lemma 8.4.1
For all $x \in \mathbb{R}$, the series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges.

Definition 8.4.2 (The Exponential Function)
Defining the exponential function $e^x : \mathbb{R} \to \mathbb{R}$ by

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Proposition 8.4.3 (Property of the Exponential)
The following are true regarding the exponential function.
- $e^{x+y} = e^x e^y$ for all $x, y \in \mathbb{R}$.
- $e^x \geq 1 + x$ for all x.
- $e^x \leq \frac{1}{1-x}$ for all $x < 1$.
- e^x is strictly increasing.
- $\frac{d}{dx}e^x = e^x$.

Proof
Let $x, y \in \mathbb{R}$.

- We want to show that

$$\sum_{k=0}^{2m} \frac{(x+y)^k}{k!} - \left(\sum_{i=0}^{m} \frac{x^i}{i!}\right)\left(\sum_{j=0}^{m} \frac{x^j}{j!}\right) \to 0$$

This is equivalent to saying that $e^{x+y} - e^x e^y = 0$. From the binomial theorem, we have

$$(x+y)^k = \sum_{i=0}^{k} \frac{k!}{i!(k-i)!} x^i y^{k-i} = \sum_{i+j=k} k! \frac{x^i}{i!} \frac{y^j}{j!}$$

Hence

$$\begin{aligned}
&\sum_{k=0}^{2m} \frac{(x+y)^k}{k!} - \left(\sum_{i=0}^{m} \frac{x^i}{i!}\right)\left(\sum_{j=0}^{m} \frac{x^j}{j!}\right) \\
&= \sum_{i+j=k} k! \frac{x^i}{i!} \frac{y^j}{j!} - \left(\sum_{i=0}^{m} \frac{x^i}{i!}\right)\left(\sum_{j=0}^{m} \frac{x^j}{j!}\right) \\
&= \sum_{i \geq m+1, i+j \leq 2m} \frac{x^i}{i!} \frac{y^j}{j!} \\
&\leq \sum_{i \geq m+1, i+j \leq 2m} \frac{|x^i|}{i!} \frac{|y^j|}{j!} \\
&= \sum_{i \geq m+1, j \geq 0} \frac{|x^i|}{i!} \frac{|y^j|}{j!} \\
&= \left(\sum_{i \geq m+1} \frac{|x^i|}{i!}\right)\left(\sum_{j=0}^{\infty} \frac{|y^j|}{j!}\right)
\end{aligned}$$

The first sum tends to 0 as m tends to infinity.

- When $x \geq 0$, $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \geq 1 + x$
- When $0 \leq x < 1$, we have $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \leq 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$. Now suppose that $x = -u$ is negative, then

$$e^u \geq 1 + u$$

thus $e^{-x} \geq 1 - x$. This implies that $\frac{1}{1-x} \geq e^x$ thus this equality is true for all $x < 1$. Now to prove the previous inequality, suppose $x \leq -1$, then $e^x > 0 > 1 + x$. Now suppose that $-1 < x < 0$. Let $x = -u$ then $0 < u < 1$ hence $e^u \leq \frac{1}{1-u}$. This implies that $e^{-x} \leq \frac{1}{1+x}$ thus

$$1 + x \leq e^x$$

- Suppose $x < y$. Then we have

$$\begin{aligned}
e^y &= e^{y-x} e^x \\
&\geq (1 + y - x) e^x \\
&> e^x
\end{aligned}$$

- Clearly differentiating e^x term by term gives e^x.

■

Definition 8.4.4 (The Logarithm)
Define the logarithm function as the inverse of e^x as $\ln(x)$.

Proposition 8.4.5
The following are true regarding the logarithm function.

- $\ln(x) : (0, \infty) \to \mathbb{R}$ is continuous.
- $\ln(x)$ is differentiable and $\frac{d}{dx}\ln(x) = \frac{1}{x}$.

Proof
Since e^x is continuous then $\ln(x)$ is also continuous. By theorem 5.2.2, $\ln(x)$ is differentiable. Also by theorem 5.2.2 we have $\frac{d}{dx}\ln(x) = \frac{1}{x}$. ■

Proposition 8.4.6
For all $-1 \leq x < 1$, we have

$$\ln(1-x) = -\sum_{k=1}^{\infty} \frac{x^k}{k}$$

Be cautious that the taylor series is only defined for $x \in [-1, 1)$. Outside of the interval, the power series will diverge. With that in mind, we will complete the chapter by defining real exponents. This is defined through both the exponential function and the logarithm.

Definition 8.4.7 (Real Exponents)
Let $x \in \mathbb{R}$ and $p \in \mathbb{R}$. Define the x to the power p to be the real number

$$x^p = e^{p\ln(x)}$$

Proposition 8.4.8 (Law of Exponents)
Let $x > 0$ and $p, q \in \mathbb{R}$.
- $x^{p+q} = x^p x^q$
- $(x^p)^q = x^{pq}$

Proof
Simply revert the real exponents in terms of e. ■